

Prime-power congruences for level-one eigenforms

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Abstract

We study prime-power congruences for the prime-to- p Hecke eigensystems of normalized cuspidal level-one eigenforms as the weight varies, and we address the problem of classifying such systems for small prime powers. We construct Hecke-equivariant transfer maps between level-one modular-symbol modules via multiplication by suitable lifts of Dickson invariants; these maps reduce the problem to finite computation and propagate Hecke operator relations through all weights. For $p = 2, 3, 5, 7$, we obtain all-weight classifications modulo 2^8 , 3^3 , 5^2 , and 7^2 , respectively.

Contents

I	General theory	3
1	Introduction and main results	3
1.1	The problem	3
1.2	Congruence conventions	5
1.3	Cyclotomic packets and twisted Eisenstein congruences	6
1.4	Main results	7
2	Modular forms and Hecke algebras	10
2.1	Hecke eigensystems, coefficient quotients, and completed Hecke algebras . .	10
2.2	Lifting shallow Hecke generators	11
2.2.1	$p = 2, 3$	12
2.2.2	$p = 5$	13
2.2.3	$p = 7$	14
2.3	Weak realization of cyclotomic packets	15
3	Modular symbols of level 1	17
3.1	Modular symbols with coefficients in R	17
3.2	Hecke operators	18
3.3	Comparison with modular forms	18
3.4	Dickson multipliers	19

3.5	Transfer map for modular symbols	21
3.6	For odd p	22
3.7	Propagation of Hecke operator relations	23
3.8	Calculus of linear relations	26
3.9	Propagation of divided Hecke relations	30
3.10	Recursive computation of Manin quotients and ideal images	32
II	Computations and classifications	35
4	The prime 2: classification modulo 256	35
4.1	Certified Hecke operator identities	35
4.2	Congruences for a_3 and a_5	36
4.3	Packet selection	37
4.4	Final classification	38
5	The prime 3: classification modulo 27	38
5.1	Certified Hecke operator identities	39
5.1.1	The nonzero branch	39
5.1.2	The zero branch	40
5.2	Congruences for a_2 and a_7	41
5.2.1	The nonzero branch	41
5.2.2	The zero branch	42
5.3	Packet selection	43
5.4	Final classification	43
6	The prime 5: classification modulo 25	44
6.1	Certified Hecke operator identities	44
6.2	Congruences for a_2 and a_{19}	46
6.3	Packet selection	47
6.4	Final classification	47
7	The prime 7: classification modulo 49	48
7.1	Certified Hecke operator identities	48
7.2	Congruences for a_3 and a_{29}	50
7.3	Packet selection	54
7.4	Final classification	55
III	Applications	56

8 Strong prime-to- p weight bounds	56
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A Signatures modulo 49	59
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Part I

General theory

1 Introduction and main results

1.1 The problem

In this work, we study congruences modulo prime powers between systems of Hecke eigenvalues arising from normalized cuspidal eigenforms of level 1. We will use the adjective *strong*, following [CKW13, KRW16], to refer to either eigenforms in characteristic 0, or Hecke eigensystems arising from such eigenforms. More precisely, for a fixed prime power p^m , we ask which strong prime-to- p Hecke eigensystems can occur in level 1 as the weight varies, and we seek an explicit classification of these systems.

Questions of this kind belong to the general study of modular forms and Galois representations modulo prime powers. For fixed level N , prime $p \nmid N$, and integer $m \geq 1$, the author, in joint work with Kiming and Wiese, formulated the following finiteness conjecture [KRW16, Conjecture 1].

Conjecture 1.1. There are only finitely many modulo p^m congruence classes of strong Hecke eigenforms on $\Gamma_1(N)$.

Equivalently, there should exist a constant $B(N, p, m)$, called a *strong weight bound*, such that every normalized characteristic-zero eigenform f on $\Gamma_1(N)$ is congruent modulo p^m to a normalized eigenform g on $\Gamma_1(N)$ of weight at most $B(N, p, m)$. A corresponding finiteness statement in characteristic p is known: for fixed level, only finitely many systems of Hecke eigenvalues with values in $\overline{\mathbf{F}}_p$ occur as the weight varies [Joc82].

The case of level 1 already exhibits much of the subtlety of the general problem. Classical results of Hatada [Hat77a, Hat77b, Hat79, Hat81] established a series of striking congruences for Hecke eigenvalues. However, these results generally concern individual Hecke eigenvalues and do not determine whole eigensystems. We record a few representative examples. For every normalized level-one eigenform f , Hatada proved, among other congruences,

$$a_\ell(f) \equiv 1 + \ell \pmod{2^4 \overline{\mathbf{Z}}_2} \quad (\ell \equiv \pm 1 \pmod{2^3}),$$

and

$$a_\ell(f) \equiv 1 + \ell \pmod{2^5 \overline{\mathbf{Z}}_2} \quad (\ell \equiv \pm 1 \pmod{2^4})$$

[Hat79, Hat81].

Coleman and Stein [CS04, Conjecture 3.4] conjectured that exactly five systems of Hecke eigenvalues modulo 3^2 occur in levels 1, 3, and 9, all first occurring at level 1, and gave computational evidence for this conjecture.

Calegari and Emerton subsequently introduced a cohomological approach which allowed them to reprove and extend several of Hatada's congruences [CE04, Theorem 2.2 and Lemmas 3.1–3.3].

For $p = 7$, Hatada had recorded a conjecture of Serre predicting

$$a_\ell(f) \equiv 1 + \ell \pmod{7\overline{\mathbf{Z}}_7} \quad (\ell \equiv \pm 1 \pmod{7}). \quad (1.1)$$

Calegari and Emerton proved the weaker congruence

$$a_\ell(f) \equiv 1 + \ell \pmod{7^{1/2}\overline{\mathbf{Z}}_7} \quad (\ell \equiv \pm 1 \pmod{7})$$

[CE04, Lemma 3.2]. They also proved, for $p = 3$,

$$a_\ell(f) \equiv 1 + \ell \pmod{3^2\overline{\mathbf{Z}}_3} \quad (\ell \equiv \pm 1 \pmod{3^2}),$$

for every normalized level-one eigenform f [CE04, Lemma 3.3].

In [Rus19], the author obtained a stronger result at level 1. For $p = 2$, all systems of Hecke eigenvalues modulo 2^7 were determined; in particular, for every normalized level-one eigenform f of weight k and every odd prime ℓ ,

$$a_\ell(f) \equiv 1 + \ell^{k-1} \pmod{2^7\overline{\mathbf{Z}}_2}.$$

This gave further evidence for Conjecture 1.1.

There is also a methodological continuity between these works. Calegari and Emerton point out that modular symbols are particularly well adapted to proving congruences that are independent of the weight, and interpret Hatada's calculations in this framework [CE04]. Their approach gives a cohomological formulation of the same general idea. The method of [Rus19] was also in this spirit, using Merel's explicit theory of modular symbols to obtain finite computations from which congruences could be deduced.

The present work continues this approach and develops it further. In particular, we are able to prove the level 1 cases of Serre's conjecture (1.1) and Coleman-Stein's conjecture [CS04, Conjecture 3.4]. The main new ingredient is a family of Hecke-equivariant transfer maps entirely within level 1, obtained by multiplication by suitable lifts of Dickson invariants. These maps propagate relations between level-one modular-symbol modules of different degrees and thereby reduce all-weight congruence questions to finite computation. In this way, we first obtain congruences for selected Hecke eigenvalues. Then we show that the Hecke operators to which these eigenvalues correspond generate the completed p -adic Hecke algebra, so that these congruences are enough to control every other Hecke eigenvalue. We do this by lifting Medvedovsky's [Med15] explicitly determined generators of the corresponding mod p Hecke algebras.

In this sense, the methods used here may be viewed as a more systematic development of the ideas introduced in [Rus19]. We do not claim that the results obtained here are the limits of what one could prove using this method: the necessary computations were run on the author's modest laptop, and we believe that more can be achieved on a more powerful machine.

In the GitHub repository [Rus26], we provide Python/SageMath code, interfacing with PARI/GP, which can be used to run and verify the computational work needed for the results of this paper. We also provide an alternative backend, written in Nim and interfacing with FLINT, which computes the needed spaces of modular symbols and the actions of the Hecke operators. The two routes are theoretically identical, but the Nim code is much faster, and can be used to verify the computationally heavy case of mod 7^2 classification. For each of our main theorems, we present the verification through Jupyter notebooks which can be run on a Sage server.

In the next few introductory subsections, we fix notation and state the main results of the paper.

Acknowledgements and AI use disclosure. The core of this work was developed by the author while employed as Assistant Professor at Koç University, Istanbul, during 2018–2020. In particular, the level 1 Coleman-Stein conjecture, and the level 1 Serre conjecture modulo $7^{3/4}\mathbf{Z}_7$, were obtained at that time.

AI tools, in particular Codex/GPT-5.6 Sol, were used to extend these initial results. AI contributions consisted mainly in orchestrating the identification of the conjectural classifications and weight bounds in each case. AI also contributed significantly to the search for the modular-symbol Hecke operator identities used in Part II to obtain congruences for selected Hecke eigenvalues. As the working modulus increases, these identities become increasingly complicated, and AI was useful for rapidly testing, rejecting, and identifying plausible identities.

All computational results obtained using AI-generated code were subsequently verified independently with code written by the author. AI was also used for proofreading the manuscript, correcting typographical errors, and identifying notational inconsistencies. All mathematical statements, proofs, computations, and references in the final manuscript were independently checked by the author, who takes full responsibility for the contents of the paper.

1.2 Congruence conventions

Let p be a prime number, and fix once and for all algebraic closures $\overline{\mathbf{Q}}$, with ring of integers $\overline{\mathbf{Z}}$, and $\overline{\mathbf{Q}}_p$, with ring of integers $\overline{\mathbf{Z}}_p$, together with an embedding

$$\overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_p.$$

Let v_p denote the normalized valuation on $\overline{\mathbf{Q}}_p$, so that $v_p(p) = 1$.

We first clarify what we mean by congruence modulo p^m , since two different notions arise when the coefficient field is ramified.

Definition 1.2. Let $\mathcal{O}$ be the ring of integers of a finite extension of $\mathbf{Q}_p$, let π be a uniformizer, and let $e_{\mathcal{O}}$ be the ramification index, so that $p\mathcal{O} = \pi^{e_{\mathcal{O}}}\mathcal{O}$. For $m \geq 1$ and $x, y \in \mathcal{O}$, define

$$\begin{aligned} x \equiv_{\text{val}} y \pmod{p^m} &\iff v_p(x - y) > m - 1, \\ x \equiv_{\text{lit}} y \pmod{p^m} &\iff x - y \in p^m\mathcal{O}. \end{aligned}$$

We call the first a *valuative congruence modulo p^m* and the second a *literal congruence modulo p^m* . The valutive convention is the one used in [KRW16, §1.1, pp. 479–480].

Equivalently, valutive congruence modulo p^m is equality in $\mathcal{O}/\pi^{e_{\mathcal{O}}(m-1)+1}\mathcal{O}$, whereas literal congruence modulo p^m is equality in $\mathcal{O}/p^m\mathcal{O} = \mathcal{O}/\pi^{me_{\mathcal{O}}}\mathcal{O}$. If the coefficient field is unramified, the two notions agree. If $e_{\mathcal{O}} > 1$, literal congruence is stronger. More generally, literal congruence modulo p^m implies valutive congruence modulo p^m , while valutive congruence modulo p^{m+1} implies literal congruence modulo p^m . Indeed,

$$\pi^{e_{\mathcal{O}}m+1}\mathcal{O} \subseteq \pi^{me_{\mathcal{O}}}\mathcal{O} \subseteq \pi^{e_{\mathcal{O}}(m-1)+1}\mathcal{O}.$$

For modular forms, congruence is understood coefficientwise. Thus, if

$$f = \sum_{n \geq 0} a_n(f)q^n, \quad g = \sum_{n \geq 0} a_n(g)q^n$$

have coefficients in $\mathcal{O}$, we write

$$f \equiv g \pmod{p^m}$$

valuatively, respectively literally, if

$$a_n(f) \equiv a_n(g) \pmod{p^m} \quad (n \geq 0)$$

in the valutive, respectively literal, sense. The Fourier coefficients of a normalized Hecke eigenform f (normalized meaning that $a_1(f) = 1$) are algebraic integers; hence, under our fixed embedding

$$\overline{\mathbf{Q}} \hookrightarrow \overline{\mathbf{Q}}_p,$$

they belong to $\overline{\mathbf{Z}}_p$. If f and g are normalized Hecke eigenforms, we say that their *prime-to- p Hecke eigensystems* are congruent modulo p^m , valuably or literally, if

$$a_n(f) \equiv a_n(g) \pmod{p^m} \quad ((n, p) = 1)$$

in the corresponding sense. Such congruences may be checked in the ring of integers of any finite extension of $\mathbf{Q}_p$ containing the coefficients of both forms.

1.3 Cyclotomic packets and twisted Eisenstein congruences

Let φ denote the Euler totient function.

Definition 1.3 (Cyclotomic packet). Let

$$a, b \in \mathbf{Z}/\varphi(p^m)\mathbf{Z}.$$

Define

$$\text{Cyc}_{a,b} : \{n \in \mathbf{Z}_{\geq 1} : (n, p) = 1\} \longrightarrow \mathbf{Z}/p^m\mathbf{Z}$$

as follows. For n prime to p , choose nonnegative integer representatives of a and b , and put

$$\text{Cyc}_{a,b}(n) := \prod_{\ell^s \parallel n} \sum_{j=0}^s \ell^{a(s-j)+bj} \pmod{p^m}.$$

This residue class is independent of the chosen representatives of a and b . We call $\text{Cyc}_{a,b}$ the *cyclotomic packet* with exponents (a, b) . At primes $\ell \neq p$,

$$\text{Cyc}_{a,b}(\ell) = \ell^a + \ell^b.$$

Moreover,

$$\text{Cyc}_{a,b} = \text{Cyc}_{b,a},$$

so the packet depends only on the unordered pair $\{a, b\}$.

Let $\theta := q \frac{d}{dq}$ denote the usual operator on q -expansions. Thus

$$\theta^r \left(\sum_{n \geq 0} a_n q^n \right) = \sum_{n \geq 1} n^r a_n q^n \quad (r \geq 1).$$

For even $\kappa \geq 4$, let

$$G_\kappa := -\frac{B_\kappa}{2\kappa} + \sum_{n=1}^{\infty} \sigma_{\kappa-1}(n) q^n, \quad \sigma_{\kappa-1}(n) := \sum_{d|n} d^{\kappa-1},$$

with B_κ the κ -th Bernoulli number, be the level 1 Eisenstein series of weight κ . For $i \geq 1$, the constant term of G_κ is killed by θ^i , and all remaining coefficients are integral. Thus

$$\theta^i G_\kappa \pmod{p^m}$$

is well defined as a q -expansion.

Definition 1.4 (Cyclotomic Hecke eigensystem). Let f be a modular form with coefficients in $\bar{\mathbf{Z}}_p$, let $e \in \mathbf{Z}_{\geq 0}$, and let $i \in \mathbf{Z}_{\geq 1}$. If

$$\theta^e f \equiv \theta^i G_\kappa \pmod{p^m},$$

we will refer to the Hecke eigensystem arising from f as *cyclotomic*. As in Definition 1.2, the congruence may be understood valuatively or literally.

To explain this terminology, note that for $(n, p) = 1$, one has

$$n^i \sigma_{\kappa-1}(n) = \text{Cyc}_{i, i+\kappa-1}(n) \quad \text{in } \mathbf{Z}/p^m \mathbf{Z}.$$

Consequently, if e is a positive multiple of $\varphi(p^m)$ and $e, i \geq m$, then, in either the valutive or literal sense,

$$\theta^e f \equiv \theta^i G_\kappa \pmod{p^m}$$

is equivalent to

$$a_n(f) \equiv \text{Cyc}_{i, i+\kappa-1}(n) \pmod{p^m} \quad ((n, p) = 1).$$

In particular, at a prime $\ell \neq p$,

$$a_\ell(f) \equiv \ell^i + \ell^{i+\kappa-1} \pmod{p^m}.$$

1.4 Main results

We now state the main results of the paper. Throughout this subsection, let f be a normalized cuspidal level-one eigenform of even weight k , and let $\mathcal{O}_f$ be the ring of integers of a finite extension of $\mathbf{Q}_p$ containing the coefficients of f .

Theorem 1.5 (Prime-to-2 classification modulo 2^8). *If*

$$k - 2 \equiv 2, 4, 10, 12 \pmod{16},$$

then

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{0, k-1}(n) \pmod{2^8} \quad ((n, 2) = 1).$$

If

$$k - 2 \equiv 0, 6, 8, 14 \pmod{16},$$

then exactly one of

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{0, k-1}(n) \pmod{2^8} \quad ((n, 2) = 1)$$

and

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{16, k-17}(n) \pmod{2^8} \quad ((n, 2) = 1)$$

holds.

Equivalently, in the first case,

$$\theta^{64}f \equiv_{\text{val}} \theta^{64}G_k \pmod{2^8},$$

whereas in the second case exactly one of

$$\theta^{64}f \equiv_{\text{val}} \theta^{64}G_k \pmod{2^8}$$

and

$$\theta^{64}f \equiv_{\text{val}} \theta^{16}G_{k+32} \pmod{2^8}$$

holds.

Conversely, every cyclotomic packet permitted above is realized as a strong prime-to-2 Hecke eigensystem modulo 2^8 in level 1. More precisely, for every even residue class $k_0 \pmod{64}$, each permitted packet is realized by a normalized cuspidal level-one strong eigenform of some weight

$$K \equiv k_0 \pmod{64}.$$

Theorem 1.6 (Prime-to-3 classification modulo 3^3). *There exists a unique unordered pair*

$$\{a, b\} \subset \mathbf{Z}/18\mathbf{Z}, \quad a + b \equiv k - 1 \pmod{18},$$

such that

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(n) \pmod{3^3} \quad ((n, 3) = 1).$$

The pair belongs to the following list:

$k \pmod{18}$	$\{a, b\}$
0	$\{0, 17\}, \{8, 9\}$
2	$\{0, 1\}, \{9, 10\}$
4	$\{0, 3\}, \{6, 15\}, \{9, 12\}$
6	$\{0, 5\}, \{9, 14\}$
8	$\{0, 7\}, \{9, 16\}$
10	$\{0, 9\}, \{3, 6\}, \{12, 15\}$
12	$\{0, 11\}, \{2, 9\}$
14	$\{0, 13\}, \{4, 9\}$
16	$\{0, 15\}, \{3, 12\}, \{6, 9\}.$

Equivalently, the pair $\{a, b\}$ can be written

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/18\mathbf{Z}$$

for some $i \geq 3$ and some even integer $\kappa \geq 4$, and

$$\theta^{18}f \equiv_{\text{val}} \theta^i G_\kappa \pmod{3^3}.$$

Conversely, every possible such cyclotomic packet is realized as a strong prime-to-3 Hecke eigensystem modulo 3^3 in level 1. If

$$k \equiv 0, 2 \pmod{6},$$

then these congruences are in fact literal.

Theorem 1.6 implies the Coleman-Stein conjecture at level 1.

Theorem 1.7 (Prime-to-5 classification modulo 5^2). *There exists a unique unordered pair*

$$\{a, b\} \subset \mathbf{Z}/20\mathbf{Z}, \quad a + b \equiv k - 1 \pmod{20},$$

such that

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(n) \pmod{5^2} \quad ((n, 5) = 1).$$

The pair belongs to the following list:

$k \bmod 20$	$\{a, b\}$
0	$\{0, 19\}, \{3, 16\}, \{4, 15\}, \{5, 14\}, \{6, 13\}, \{9, 10\}$
2	$\{0, 1\}, \{4, 17\}, \{5, 16\}, \{6, 15\}, \{7, 14\}, \{10, 11\}$
4	$\{0, 3\}, \{4, 19\}, \{5, 18\}, \{8, 15\}, \{9, 14\}, \{10, 13\}$
6	$\{0, 5\}, \{2, 3\}, \{6, 19\}, \{9, 16\}, \{10, 15\}, \{12, 13\}$
8	$\{0, 7\}, \{1, 6\}, \{2, 5\}, \{10, 17\}, \{11, 16\}, \{12, 15\}$
10	$\{0, 9\}, \{1, 8\}, \{4, 5\}, \{10, 19\}, \{11, 18\}, \{14, 15\}$
12	$\{0, 11\}, \{1, 10\}, \{2, 9\}, \{5, 6\}, \{12, 19\}, \{15, 16\}$
14	$\{0, 13\}, \{3, 10\}, \{4, 9\}, \{5, 8\}, \{14, 19\}, \{15, 18\}$
16	$\{0, 15\}, \{1, 14\}, \{4, 11\}, \{5, 10\}, \{7, 8\}, \{17, 18\}$
18	$\{0, 17\}, \{1, 16\}, \{2, 15\}, \{5, 12\}, \{6, 11\}, \{7, 10\}.$

Equivalently, writing the selected pair as

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/20\mathbf{Z}$$

with $i \geq 2$ and an even integer $\kappa \geq 4$, one has

$$\theta^{20} f \equiv_{\text{val}} \theta^i G_\kappa \pmod{5^2}.$$

Conversely, every packet in the table is realized as a strong prime-to-5 Hecke eigensystem modulo 5^2 in level 1.

Theorem 1.8 (Prime-to-7 classification modulo 49). *There exists a unique triple*

$$(c, x, y) \in \{0, \dots, 6\}^3$$

listed for the weight residue $k \bmod 42$ in Table 2 of Appendix A, such that

$$a_3(f) \equiv_{\text{val}} c + 7x \pmod{49}, \quad a_{29}(f) \equiv_{\text{val}} 2 + 7y \pmod{49}.$$

Two normalized cuspidal level-one eigenforms have congruent prime-to-7 Hecke eigensystems modulo 49, in the valuative sense, if and only if their weights are congruent modulo 42 and their corresponding triples (c, x, y) agree.

Conversely, every signature in Table 2 is realized by a normalized cuspidal level-one eigenform of weight at most 380 in the corresponding residue class modulo 42. Thus there are exactly 315 prime-to-7 Hecke eigensystems modulo 49, in the valuative sense.

Modulo 7, in the literal sense, these eigensystems are cyclotomic. More precisely, there exists a unique unordered pair

$$\{a, b\} \subset \mathbf{Z}/6\mathbf{Z}, \quad a + b \equiv k - 1 \pmod{6},$$

such that

$$a_n(f) \equiv_{\text{lit}} \text{Cyc}_{a,b}(n) \pmod{7} \quad ((n, 7) = 1).$$

The pair belongs to the following list:

$k \bmod 6$	$\{a, b\}$
0	$\{0, 5\}, \{1, 4\}, \{2, 3\}$
2	$\{0, 1\}, \{2, 5\}, \{3, 4\}$
4	$\{0, 3\}, \{1, 2\}, \{4, 5\}$.

Equivalently, the pair may be written

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/6\mathbf{Z}$$

for some

$$i \in \{1, \dots, 6\}, \quad \kappa \in \{4, 8\}, \quad \kappa + 2i \equiv k \pmod{6},$$

and

$$\theta^6 f \equiv_{\text{lit}} \theta^i G_\kappa \pmod{7}.$$

Theorem 1.8 implies Serre's conjecture at level 1.

2 Modular forms and Hecke algebras

2.1 Hecke eigensystems, coefficient quotients, and completed Hecke algebras

Denote by $S_k(\mathbf{Z}) \subset \mathbf{Z}[[q]]$ the lattice generated by the q -expansions of level 1 cuspforms of weight k with coefficients in $\mathbf{Z}$. Let K be a finite extension of $\mathbf{Q}_p$ with ring of integers $\mathcal{O}$. We will denote by $S_k(\mathcal{O})$ the $\mathcal{O}$ -submodule of $\mathcal{O}[[q]]$ spanned by the image of $S_k(\mathbf{Z})$.

For $h \geq 0$, put

$$S_{\leq h}(\mathcal{O}) := \sum_{0 \leq k \leq h} S_k(\mathcal{O}), \quad D_{\leq h}(\mathcal{O}) := (S_{\leq h}(\mathcal{O}) \otimes_{\mathcal{O}} K) \cap \mathcal{O}[[q]].$$

Equivalently, $D_{\leq h}(\mathcal{O})$ is the p -power saturation of $S_{\leq h}(\mathcal{O})$ inside $\mathcal{O}[[q]]$. For B a p -adically complete and separated local $\mathcal{O}$ -algebra B , put

$$D_{\leq h}(B) := D_{\leq h}(\mathcal{O}) \otimes_{\mathcal{O}} B$$

and denote by $S_k(B)$ the image of $S_k(\mathcal{O})$ in $D_{\leq h}(B)$. These spaces inherit the action of the Hecke operators from the spaces of modular forms.

We recall the definition of strong, weak, and dc-weak eigenform, first introduced in [CKW13].

Definition 2.1 (Strong, weak, and dc-weak eigenforms). Let B be a finite local $\mathbf{Z}_p$ -algebra and let $f = q + \sum_{n \geq 2} a_n(f)q^n \in B[[q]]$ be a normalized q -expansion with coefficients in B .

1. The form f is *strong* if it is obtained by reducing a normalized characteristic-zero eigenform through a coefficient-ring map to B .
2. The form f is *weak* if $f \in S_k(B)$ for some k and is a simultaneous Hecke eigenform there.
3. The form f is *dc-weak* if $f \in D_{\leq h}(B)$ for some h and is a simultaneous Hecke eigenform there.

Definition 2.2. For each h , define

$$\mathbb{T}_{\leq h}^{(p)}(R) := \text{im}(R[\mathbb{T}_n : (n, p) = 1] \rightarrow \text{End}_R(D_{\leq h}(R))). \quad (2.1)$$

If $h \leq h'$, restriction from $D_{\leq h'}(R)$ to $D_{\leq h}(R)$ allows us to define the completed algebra

$$\mathbb{T}_{\text{dc}}^{(p)}(R) := \varprojlim_h \mathbb{T}_{\leq h}^{(p)}(R). \quad (2.2)$$

We will usually write $\mathbb{T}_{\text{dc}}^{(p)} := \mathbb{T}_{\text{dc}}^{(p)}(\mathbf{Z}_p)$.

For $u \in \mathbf{Z}_p^\times$, Katz's weight action $[u]$ on divided congruences acts on weight k by u^k ; see [Kat75a, (2.4)] and [Kat75b, §X]. The extension to the low-level cases $N < 3$ used here is explained in [Rus19, Theorem 2.2 and the following paragraph]. For odd p , put $t_\Lambda := [1 + p]$, and for $p = 2$, put $t_\Lambda := [5]$. It can be shown that $t_\Lambda \in \mathbb{T}_{\text{dc}}^{(p)}(R)$ [BK15, Lemma 9].

Hecke eigensystems away-from- p arising from strong, weak, or dc-weak eigenforms with coefficients in R correspond to continuous $\mathbf{Z}_p$ -algebra homomorphisms $\mathbb{T}_{\text{dc}}^{(p)} \rightarrow R$.

Definition 2.3 (Shallow Hecke algebra). Let $M_{\leq h}(\mathbf{F}_p)$ be the space of classical level-one modular forms modulo p of weights at most h . Following the terminology of [BK15, §1.2], define the *shallow Hecke algebra* by

$$\mathbb{T}^{\text{sh},(p)} := \varprojlim_h \text{im}(\mathbf{F}_p[\mathbb{T}_n : (n, p) = 1] \longrightarrow \text{End}_{\mathbf{F}_p}(M_{\leq h}(\mathbf{F}_p))).$$

For a residual prime-to- p eigensystem τ , its completed local factor is denoted $\mathbb{T}_\tau^{\text{sh},(p)}$.

2.2 Lifting shallow Hecke generators

Let $\mathbf{F}/\mathbf{F}_p$ be a finite extension, and put

$$\mathbb{T}_{\mathbf{F}}^{\text{sh},(p)} := \mathbb{T}^{\text{sh},(p)} \otimes_{\mathbf{F}_p} \mathbf{F}.$$

For an $\mathbf{F}$ -valued residual eigensystem

$$\tau : \mathbb{T}_{\mathbf{F}}^{\text{sh},(p)} \longrightarrow \mathbf{F},$$

let

$$\mathfrak{m}_\tau^{\text{sh}} := \ker(\tau),$$

and denote by $\mathbb{T}_\tau^{\text{sh},(p)}$ the corresponding completed local factor.

Let $\mathcal{O}$ be the ring of integers of a finite extension of $\mathbf{Q}_p$ with residue field $\mathbf{F}$, and assume that all residual prime-to- p eigensystems are $\mathbf{F}$ -valued. The Hecke algebra $\mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O})$ is a complete Noetherian semilocal ring, with decomposition

$$\mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O}) = \prod_{\tau} \mathbb{T}_{\text{dc},\tau}^{(p)}(\mathcal{O}),$$

where τ runs over the residual prime-to- p eigensystems.

Lemma 2.4 (Lifting shallow generators). *Assume that $p \geq 5$. Let*

$$\mathcal{H} = \{h_1, \dots, h_s\} \subseteq \mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O}),$$

and write $\bar{h}_i$ for the image of h_i in $\mathbb{T}_{\mathbf{F}}^{\text{sh},(p)}$. Suppose that:

1. for every residual eigensystem τ , the classes of

$$\bar{h}_i - \tau(\bar{h}_i), \quad 1 \leq i \leq s,$$

span

$$\mathfrak{m}_\tau^{\text{sh}} / (\mathfrak{m}_\tau^{\text{sh}})^2;$$

2. the map

$$\tau \longmapsto (\tau(\bar{h}_1), \dots, \tau(\bar{h}_s)) \in \mathbf{F}^s$$

is injective.

Then

$$\mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O}) = \overline{\mathcal{O}[t_\Lambda, h_1, \dots, h_s]}.$$

Proof. The passage from generators of

$$\mathfrak{m}_\tau^{\text{sh}} / (\mathfrak{m}_\tau^{\text{sh}})^2$$

to topological generators of the completed local factor $\mathbb{T}_{\text{dc}, \tau}^{(p)}(\mathcal{O})$, using Nakayama's lemma, is the argument used in [Rus19, §7]. We recall it briefly. By [BK15, Lemma 9], $t_\Lambda = [1 + p]$ belongs to $\mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O})$, and the argument in the proof of [BK15, Proposition 16] gives, after extension of scalars and passage to the τ -local factor,

$$\mathbb{T}_{\text{dc}, \tau}^{(p)}(\mathcal{O}) / (\pi, t_\Lambda - 1) \mathbb{T}_{\text{dc}, \tau}^{(p)}(\mathcal{O}) \simeq \mathbb{T}_\tau^{\text{sh}, (p)}.$$

Hypothesis (1), together with Nakayama's lemma, therefore shows that $t_\Lambda, h_1, \dots, h_s$ topologically generate every completed local factor as an $\mathcal{O}$ -algebra.

Put $A = \overline{\mathcal{O}[t_\Lambda, h_1, \dots, h_s]} \subset \mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O})$, and let $\mathfrak{m}_\tau$ be the maximal ideal of $\mathbb{T}_{\text{dc}, \tau}^{(p)}(\mathcal{O})$. The preceding paragraph shows that A surjects onto each

$$\mathbb{T}_{\text{dc}, \tau}^{(p)}(\mathcal{O}) / \mathfrak{m}_\tau^n.$$

The maximal ideals $\mathfrak{n}_\tau := \ker(\tau|_A)$ are distinct by hypothesis (2). The kernels of these maps are therefore pairwise comaximal, so the Chinese remainder theorem shows that A surjects onto their product for every n . Thus A is dense in $\mathbb{T}_{\text{dc}}^{(p)}(\mathcal{O})$. Since A is closed by definition, the two algebras are equal. $\square$

2.2.1 $p = 2, 3$

By [Rus19, Theorem 7.2(i)–(ii) and Proposition 3.8(i)], the completed prime-to-2 Hecke algebra is

$$\mathbb{T}_{\text{dc}}^{(2)} = \overline{\mathbf{Z}_2[T_3, T_5, t_\Lambda]}, \quad t_\Lambda = [5], \tag{2.3}$$

and the completed prime-to-3 Hecke algebra is

$$\mathbb{T}_{\text{dc}}^{(3)} = \overline{\mathbf{Z}_3[T_2, T_7, t_\Lambda]}, \quad t_\Lambda = [4]. \tag{2.4}$$

Consequently, two continuous characters of $\mathbb{T}_{\text{dc}}^{(2)}$ with values in a finite $\mathbf{Z}_2$ -algebra are equal if they agree on T_3 , T_5 , and t_Λ , while two continuous characters of $\mathbb{T}_{\text{dc}}^{(3)}$ with values in a finite $\mathbf{Z}_3$ -algebra are equal if they agree on T_2 , T_7 , and t_Λ .

2.2.2 $p = 5$

Medvedovsky shows that the shallow Hecke algebra at $p = 5$ has four twist-equivalent residual local factors. One of them is

$$\tau_0 := \omega^{-1} + 1$$

where ω is the mod-5 cyclotomic character. Then

$$\mathbb{T}_{\tau_0}^{\text{sh},(5)} \simeq \mathbf{F}_5[[x, y]], \quad x := T_{11} - 2, \quad y := T_{19};$$

see [Med15, Theorem 8.1 and §8.5]. We would like to lift this result to obtain generators of $\mathbb{T}_{\text{dc}}^{(5)}$. The difficulty is that the residual values of T_{11} do not separate the four local factors. We will show that T_{11} may be replaced by T_2 .

Let Δ denote the modular discriminant of weight 12, and let E_4 denote the normalized Eisenstein series of weight 4. Consider the following forms modulo 5:

$$\begin{aligned} f_0 &= 2\Delta + \Delta^2, \\ f_1 &= 3\Delta + 4\Delta^3 + \Delta^4, \\ f_2 &= 3\Delta + \Delta^3 + \Delta^5 + 3\Delta^6 + 2\Delta^7 + 4\Delta^8 + \Delta^9 \\ &\quad + 2\Delta^{11} + 2\Delta^{12} + 4\Delta^{13} + \Delta^{14}. \end{aligned}$$

These are in fact mod 5 reductions of classical modular forms, as can be seen using the identity $E_4 \equiv 1 \pmod{5}$.

The residual T_2 -values on the four local factors are

$$4, \quad 3, \quad 1, \quad 2,$$

with value 4 on τ_0 . Put $s := T_2 - 4$. Then s belongs to $\mathfrak{m}_{\tau_0}^{\text{sh}}$ and is a unit on each of the other three local factors. A direct q -expansion calculation gives

	s	x	y
f_0	0	0	0
f_1	$3f_0$	0	f_0
f_2	$3f_0$	$4f_0$	$4f_0$

(2.5)

In particular, $s^2 f_i = 0$. Since s is a unit on every residual local factor other than τ_0 , it follows that f_0, f_1, f_2 belong to the τ_0 -primary summand.

We also get from (2.5) that

$$(\mathfrak{m}_{\tau_0}^{\text{sh}})^2 f_i = 0 \quad (i = 0, 1, 2).$$

Since $\mathbb{T}_{\tau_0}^{\text{sh},(5)} \simeq \mathbf{F}_5[[x, y]]$, we have

$$\dim_{\mathbf{F}_5} \mathfrak{m}_{\tau_0}^{\text{sh}} / (\mathfrak{m}_{\tau_0}^{\text{sh}})^2 = 2.$$

Suppose

$$\beta s + \gamma y = 0, \quad \beta, \gamma \in \mathbf{F}_5,$$

is a relation in $\mathfrak{m}_{\tau_0}^{\text{sh}} / (\mathfrak{m}_{\tau_0}^{\text{sh}})^2$. Then $\beta s + \gamma y$ annihilates f_1 and f_2 . By (2.5), this forces $\beta = \gamma = 0$. Thus the classes of s and y are linearly independent, and therefore form a basis

of $\mathfrak{m}_{\tau_0}^{\text{sh}}/(\mathfrak{m}_{\tau_0}^{\text{sh}})^2$. Equivalently, the classes of $T_2 - 4$ and T_{19} form a basis of this cotangent space.

By [Med15, Proposition 2.41], twisting by the θ operator permutes the four residual local factors and their cotangent spaces. Under these permutations, the two classes above are carried, up to nonzero scalars, to the classes of

$$T_2 - c_\tau, \quad T_{19},$$

where $c_\tau = \tau(T_2)$. Hence these classes span $\mathfrak{m}_\tau^{\text{sh}}/(\mathfrak{m}_\tau^{\text{sh}})^2$ for every residual eigensystem τ . Moreover, the residual values of T_2 distinguish the four residual eigensystems. Lemma 2.4 therefore gives

$$\mathbb{T}_{\text{dc}}^{(5)} = \overline{\mathbf{Z}_5[T_2, T_{19}, t_\Lambda]}, \quad t_\Lambda = [6]. \quad (2.6)$$

2.2.3 $p = 7$

Let ω be the mod-7 cyclotomic character. Medvedovsky shows that the shallow Hecke algebra has nine residual eigensystems

$$\tau_{a,b} = \omega^a + \omega^b, \quad a \in \{0, 2, 4\}, \quad b \in \{1, 3, 5\};$$

see [Med15, Theorem 8.1 and §8.7].

Lemma 2.5. *For $\tau = \tau_{a,b}$, put*

$$c_\tau := \tau(T_3) = 3^a + 3^b \in \mathbf{F}_7.$$

Then the classes of

$$T_3 - c_\tau, \quad T_{29} - 2$$

form a basis of $\mathfrak{m}_\tau^{\text{sh}}/(\mathfrak{m}_\tau^{\text{sh}})^2$.

Proof. Put $d := b - a \in \{1, 3, 5\} \pmod{6}$. By [Med15, Corollary 7.2], we need a prime ℓ such that

$$\ell^d \not\equiv 1 \pmod{7}, \quad \ell^6 \not\equiv 1 \pmod{49}.$$

These conditions are satisfied for $\ell = 3$. Thus $T_3 - c_\tau$ will be one of our generators.

For the other generator, we use Medvedovsky's criterion [Med15, Corollary 7.8]. Put

$$E := \mathcal{O}_{\mathbf{Q}(\zeta_7)}[1/7]^\times,$$

the group of 7-units of $\mathbf{Q}(\zeta_7)$. We need classes $\alpha, \beta \in E/E^7$ lying in the ω^{1-d} - and ω^{1+d} -eigenspaces, respectively. Furthermore, we need a prime $\ell \equiv 1 \pmod{7}$ such that neither α nor β are 7th powers in $\mathbf{F}_\ell$.

For $j \in \{0, 2, 4\}$, put

$$u_j := \prod_{r=1}^6 (1 - \zeta_7^r)^{r^{-j}} \in E/E^7.$$

If $\sigma_c(\zeta_7) = \zeta_7^c$, then

$$\sigma_c(u_j) = u_j^{c^j},$$

so u_j lies in the ω^j -eigenspace of E/E^7 . For $d = 1, 3, 5$, the required pairs of eigenspace exponents are

$$(1 - d, 1 + d) = (0, 2), (4, 4), (2, 0) \pmod{6}.$$

For any prime $\mathfrak{l}$ of $\mathbf{Q}(\zeta_7)$ above 29, the reductions of u_0, u_2, u_4 in

$$\mathcal{O}_{\mathbf{Q}(\zeta_7)}/\mathfrak{l} \simeq \mathbf{F}_{29}$$

are not seventh powers. Thus we can take T_{29} as our second generator. $\square$

The level-one Hecke recursion gives

$$[3] = 3(T_3^2 - T_9), \quad (2.7)$$

so

$$[3] \in \mathbb{T}_{\text{dc}}^{(7)}.$$

We apply Lemma 2.4 with

$$\mathcal{H} = \{T_3, T_{29}, [3]\}.$$

Lemma 2.5 verifies hypothesis (1) of Lemma 2.4.

It remains to verify that the residual values of the elements of $\mathcal{H}$ distinguish the nine residual eigensystems. For $\tau = \tau_{a,b}$, we have

$$\tau(T_3) = 3^a + 3^b, \quad \tau([3]) = 3^{a+b+1}.$$

The corresponding pairs are

	$b = 1$	$b = 3$	$b = 5$
$a = 0$	(4, 2)	(0, 4)	(6, 1)
$a = 2$	(5, 4)	(1, 1)	(0, 2)
$a = 4$	(0, 1)	(3, 2)	(2, 4)

These nine pairs are distinct. Hence T_3 and $[3]$ jointly distinguish the nine residual eigensystems, so hypothesis (2) of Lemma 2.4 is satisfied.

Lemma 2.4 therefore gives

$$\mathbb{T}_{\text{dc}}^{(7)} = \overline{\mathbf{Z}_7[t_\Lambda, T_3, T_{29}, [3]]}, \quad t_\Lambda = [8].$$

Finally, t_Λ is redundant. The element 3 is a topological generator of $\mathbf{Z}_7^\times$. By continuity of the weight action, $[8]$ belongs to the closed $\mathbf{Z}_7$ -algebra generated by $[3]$. Thus

$$\mathbb{T}_{\text{dc}}^{(7)} = \overline{\mathbf{Z}_7[T_3, T_{29}, [3]]}.$$

2.3 Weak realization of cyclotomic packets

To compare a strong eigensystem with a cyclotomic packet through the completed Hecke algebra, we need to know that the packet itself is realized by a genuine weak Hecke eigensystem modulo p^m .

Lemma 2.6 (Weak realization of cyclotomic packets). *Let p be a prime, let $m \geq 1$, let $\kappa \geq 4$ be even, and let $i \geq 1$. There exist an even integer K and a normalized weak cuspidal eigenform*

$$g_{\kappa,i} \in S_K(\mathbf{Z}/p^m\mathbf{Z})$$

such that

$$K \equiv \kappa + 2i \pmod{\varphi(p^m)}$$

and

$$a_n(g_{\kappa,i}) = \text{Cyc}_{i,i+\kappa-1}(n) \quad ((n,p) = 1).$$

Equivalently, the cyclotomic packet $\text{Cyc}_{i,i+\kappa-1}$ defines a continuous character

$$\lambda_{\kappa,i} : \mathbb{T}_{\text{dc}}^{(p)} \longrightarrow \mathbf{Z}/p^m\mathbf{Z}.$$

Proof. Put

$$G_{\kappa}^* := G_{\kappa} - p^{\kappa-1}G_{\kappa} \mid V_p.$$

For $n \geq 1$, its Fourier coefficients are

$$a_n(G_{\kappa}^*) = \sum_{\substack{d|n \\ p \nmid d}} d^{\kappa-1}.$$

Serre shows that G_{κ}^* is a level-one p -adic modular form of p -adic weight κ [Ser73, §1.6]. By [Ser73, Théorème 5 and the following remark], $\theta^i G_{\kappa}^*$ is a p -adic modular form of weight $\kappa + 2i$. The constant term of G_{κ}^* need not be integral, but it is killed by θ^i , while all other coefficients of $\theta^i G_{\kappa}^*$ are integral. Hence $\theta^i G_{\kappa}^*$ is integral. As a Serre p -adic modular form, $\theta^i G_{\kappa}^*$ can be approximated by classical modular forms, so there exist an even integer K and

$$F \in M_K(\mathbf{Z}_p)$$

such that

$$F \equiv \theta^i G_{\kappa}^* \pmod{p^m}, \quad K \equiv \kappa + 2i \pmod{\varphi(p^m)}.$$

The reduction of F has zero constant term and coefficient 1 at q , and defines a normalized form

$$g_{\kappa,i} \in S_K(\mathbf{Z}/p^m\mathbf{Z}).$$

For every prime $\ell \neq p$,

$$T_{\ell} \theta^i G_{\kappa}^* = \ell^i \theta^i T_{\ell} G_{\kappa}^* = \ell^i (1 + \ell^{\kappa-1}) \theta^i G_{\kappa}^*.$$

Consequently,

$$T_{\ell} g_{\kappa,i} = \ell^i (1 + \ell^{\kappa-1}) g_{\kappa,i}.$$

Thus

$$a_{\ell}(g_{\kappa,i}) = \ell^i + \ell^{i+\kappa-1} = \text{Cyc}_{i,i+\kappa-1}(\ell).$$

The Hecke recursion at prime powers and multiplicativity then give

$$a_n(g_{\kappa,i}) = \text{Cyc}_{i,i+\kappa-1}(n) \quad ((n,p) = 1).$$

Hence $g_{\kappa,i}$ realizes the stated cyclotomic packet as a weak prime-to- p Hecke eigensystem, and therefore as a character of $\mathbb{T}_{\text{dc}}^{(p)}$. $\square$

3 Modular symbols of level 1

3.1 Modular symbols with coefficients in R

Let $d \geq 0$ be even, and put

$$\Gamma := \mathrm{SL}_2(\mathbf{Z}).$$

For now, let R be a commutative ring. Let

$$V_d(R) := \mathrm{Sym}^d(R^2)$$

be the R -module of homogeneous polynomials of degree d in X and Y . We regard $V_d(R)$ as a right $\mathbf{Z}[M_2(\mathbf{Z})]$ -module via

$$(P \mid \gamma)(X, Y) := P(aX + bY, cX + dY), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbf{Z}).$$

Let

$$\mathrm{St}_R := \mathrm{Div}^0(\mathbf{P}^1(\mathbf{Q})) \otimes_{\mathbf{Z}} R$$

be the Steinberg module over R . Equivalently, St_R is generated by symbols

$$\{\alpha, \beta\}, \quad \alpha, \beta \in \mathbf{P}^1(\mathbf{Q}),$$

subject to the relations

$$\{\alpha, \alpha\} = 0, \quad \{\alpha, \beta\} + \{\beta, \eta\} + \{\eta, \alpha\} = 0.$$

We equip St_R with the right Γ -action induced by fractional linear transformations, and

$$\mathrm{St}_R \otimes_R V_d(R)$$

with the diagonal right action. Following [Wie09, Notation 2.1 and Definition 3.1], define the *modular symbols of level 1 with coefficients in R* to be the coinvariants

$$\mathbb{M}_d(R) := H_0(\Gamma, \mathrm{St}_R \otimes_R V_d(R)).$$

Let

$$S := \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad U := \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}.$$

By [Wie09, Theorem 3.5(1)], there is an isomorphism

$$\mathbb{M}_d(R) \simeq \frac{V_d(R)}{V_d(R)(1+S) + V_d(R)(1+U+U^2)}.$$

We refer to the right-hand side as the *Manin presentation* of $\mathbb{M}_d(R)$. Finally, let

$$\iota := \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since ι normalizes Γ , its action on $\mathrm{St}_R \otimes_R V_d(R)$ induces an involution on $\mathbb{M}_d(R)$. For $\epsilon \in \{+1, -1\}$, define

$$\mathbb{M}_d^\epsilon(R) := \mathbb{M}_d(R)^{\iota=\epsilon} = \ker(\iota - \epsilon).$$

3.2 Hecke operators

For $n \geq 1$ with $(n, p) = 1$, let

$$\mathcal{H}_n := \left\{ \begin{pmatrix} a & b \\ c & \delta \end{pmatrix} \in M_2(\mathbf{Z}) : a > b \geq 0, \quad \delta > c \geq 0, \quad a\delta - bc = n \right\}.$$

By [Mer94, §3.3, Proposition 20, pp. 29–30], the element

$$\sum_{M \in \mathcal{H}_n} M \in \mathbf{Z}[M_2(\mathbf{Z})]$$

satisfies Merel's condition (C_n) . Hence [Mer94, §2.1, Theorem 4, pp. 16–17] gives an operator

$$T_n : \mathbb{M}_d(R) \longrightarrow \mathbb{M}_d(R)$$

which, in the Manin presentation of the preceding subsection, is given by

$$T_n[P] = \sum_{M \in \mathcal{H}_n} [P \mid M].$$

Here it is the sum over the Heilbronn–Merel family that descends to the Manin quotient; the individual maps $P \mapsto P \mid M$ need not do so. Although Merel formulates the Hecke theory over $\mathbf{C}$, the construction is integral: the modular-symbol construction may be carried out over an arbitrary coefficient ring, and these Hecke operators preserve the integral structure [Mer94, §1.1, p. 4 and §2.1, Remark 1, p. 18]. Thus the preceding formula defines T_n on $\mathbb{M}_d(R)$ for every commutative ring R . The operators T_n commute with the involution ι , and hence preserve $\mathbb{M}_d^\pm(R)$; see [Mer94, §2.1, Remark 2, p. 18].

3.3 Comparison with modular forms

Let

$$\mathbb{M}_d^{\text{tf}}(\mathbf{Z}) := \mathbb{M}_d(\mathbf{Z}) / \mathbb{M}_d(\mathbf{Z})_{\text{tors}}$$

be the torsion-free quotient of the integral level-one modular-symbol module. Put

$$t := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

and let $B_d^{\text{tf}}(\mathbf{Z})$ be the torsion-free quotient of

$$\frac{V_d(\mathbf{Z})}{V_d(\mathbf{Z})(t-1)}.$$

By [Wie09, Theorem 3.5(2)–(3)], the boundary map is defined integrally and induces a homomorphism

$$\partial_d^{\text{tf}} : \mathbb{M}_d^{\text{tf}}(\mathbf{Z}) \longrightarrow B_d^{\text{tf}}(\mathbf{Z}).$$

For $\epsilon \in \{+1, -1\}$, define

$$P_d^\epsilon(\mathbf{Z}) := \left(\mathbb{M}_d^{\text{tf}}(\mathbf{Z}) \right)^{\iota=\epsilon}$$

and

$$L_d^\epsilon(\mathbf{Z}) := \ker \left(\partial_d^{\text{tf}} \Big|_{P_d^\epsilon(\mathbf{Z})} \right).$$

The lattice $P_d^\epsilon(\mathbf{Z})$ is saturated in $\mathbb{M}_d^{\text{tf}}(\mathbf{Z})$. Indeed,

$$\mathbb{M}_d^{\text{tf}}(\mathbf{Z})/P_d^\epsilon(\mathbf{Z}) \simeq \text{im}(\iota - \epsilon),$$

and the latter is a subgroup of the torsion-free group $\mathbb{M}_d^{\text{tf}}(\mathbf{Z})$. The lattice $L_d^\epsilon(\mathbf{Z})$ is saturated in $P_d^\epsilon(\mathbf{Z})$, as can be seen from

$$P_d^\epsilon(\mathbf{Z})/L_d^\epsilon(\mathbf{Z}) \cong \text{im}\left(\partial_d^{\text{tf}}\Big|_{P_d^\epsilon(\mathbf{Z})}\right) \subset B_d^{\text{tf}}(\mathbf{Z}).$$

In particular, $L_d^\epsilon(\mathbf{Z})$ is saturated in $\mathbb{M}_d^{\text{tf}}(\mathbf{Z})$.

The Hecke operators preserve $P_d^\epsilon(\mathbf{Z})$ and $L_d^\epsilon(\mathbf{Z})$: they preserve the integral modular-symbol structure, commute with the involution ι , and are compatible with the boundary map; see [Mer94, §2.1, Proposition 11 and the following Remark, p. 18]. After extension of scalars to $\mathbf{C}$, the Eichler-Shimura period pairing, in the modular-symbol formulation of Merel, restricts to a nondegenerate pairing

$$S_{d+2}(\mathbf{C}) \times \left(L_d^+(\mathbf{Z}) \otimes_{\mathbf{Z}} \mathbf{C}\right) \longrightarrow \mathbf{C};$$

see [Mer94, §1.5, Theorem 3, pp. 10–11, and §1.6, Proposition 8, p. 12]. Moreover, the Hecke operators on modular forms and modular symbols are adjoint with respect to this pairing [Mer94, §2.1, Proposition 10, pp. 17–18]. Consequently, $S_{d+2}(\mathbf{C})$ and $L_d^+(\mathbf{Z}) \otimes_{\mathbf{Z}} \mathbf{C}$ carry the same systems of Hecke eigenvalues.

We now pass to local coefficients. For $m \geq 0$, let

$$R_m := \begin{cases} \mathbf{Z}_p, & m = 0, \\ \mathbf{Z}_p/p^m \mathbf{Z}_p, & m \geq 1. \end{cases}.$$

Write

$$\mathbb{M}_d^{\text{tf}}(R_m) := \mathbb{M}_d^{\text{tf}}(\mathbf{Z}) \otimes_{\mathbf{Z}} R_m, \quad P_d^\epsilon(R_m) := P_d^\epsilon(\mathbf{Z}) \otimes_{\mathbf{Z}} R_m, \quad L_d^\epsilon(R_m) := L_d^\epsilon(\mathbf{Z}) \otimes_{\mathbf{Z}} R_m.$$

When $R_m = \mathbf{Z}_p$, we will simply write $\mathbb{M}_d^{\text{tf}}$, P_d^ϵ , and L_d^ϵ . The lattices P_d^ϵ and L_d^ϵ are saturated in $\mathbb{M}_d^{\text{tf}}$ and P_d^ϵ , respectively. In particular

$$P_d^\epsilon/p^m P_d^\epsilon \hookrightarrow \mathbb{M}_d^{\text{tf}}/p^m \mathbb{M}_d^{\text{tf}}.$$

3.4 Dickson multipliers

Fix $m \geq 1$. Define

$$\alpha_p := \frac{X^{p^2}Y - XY^{p^2}}{X^pY - XY^p} = \sum_{j=0}^p X^{(p-j)(p-1)} Y^{j(p-1)}$$

and

$$\beta_p := X^pY - XY^p.$$

Remark 3.1. The choice of α_p and β_p is motivated by Dickson's theorem on invariants. In dimension 2, this theorem states that

$$\mathbf{F}_p[X, Y]^{\text{GL}_2(\mathbf{F}_p)} = \mathbf{F}_p[\bar{\alpha}_p, \bar{\beta}_p^{p-1}],$$

where bars denote reduction modulo p . See [Ste87, Theorem A, p. 699].

Define

$$A_m := \alpha_p^{p^{m-1}} \quad \text{and} \quad B_m := \beta_p^{p^{m-1}}.$$

Their degrees are

$$a_m := \deg A_m = p^m(p-1), \quad b_m := \deg B_m = p^{m-1}(p+1).$$

Lemma 3.2. *The reductions*

$$\bar{A}_m, \bar{B}_m \in \mathbf{F}_p[X, Y]$$

are coprime.

Proof. Since

$$\bar{A}_m = \bar{\alpha}_p^{p^{m-1}} \quad \text{and} \quad \bar{B}_m = \bar{\beta}_p^{p^{m-1}},$$

it is enough to prove that $\bar{\alpha}_p$ and $\bar{\beta}_p$ are coprime. Now

$$\bar{\beta}_p = XY(X^{p-1} - Y^{p-1}).$$

The polynomial $\bar{\alpha}_p$ is divisible by neither X nor Y . Moreover, we have

$$\bar{\alpha}_p \equiv (p+1)Y^{p(p-1)} = Y^{p(p-1)} \pmod{X^{p-1} - Y^{p-1}}.$$

Hence any common divisor of $\bar{\alpha}_p$ and $X^{p-1} - Y^{p-1}$ must divide $Y^{p(p-1)}$. Since

$$\gcd(X^{p-1} - Y^{p-1}, Y) = 1,$$

this common divisor is a unit. Thus

$$\gcd(\bar{\alpha}_p, \bar{\beta}_p) = 1,$$

and the result follows. □

Define the coefficient multiplication map

$$\Psi_d : V_{d-a_m}(\mathbf{Z}_p) \oplus V_{d-b_m}(\mathbf{Z}_p) \longrightarrow V_d(\mathbf{Z}_p)$$

by

$$\Psi_d(F, G) := A_m F + B_m G. \tag{3.1}$$

Proposition 3.3 (Surjective coefficient transfer). *If*

$$d \geq a_m + b_m = p^{m-1}(p^2 + 1),$$

then Ψ_d is surjective.

Proof. By Lemma 3.2, the reductions $\bar{A}_m$ and $\bar{B}_m$ are coprime in $\mathbf{F}_p[X, Y]$. Reducing Ψ_d modulo p , we obtain

$$\bar{\Psi}_d : V_{d-a_m}(\mathbf{F}_p) \oplus V_{d-b_m}(\mathbf{F}_p) \longrightarrow V_d(\mathbf{F}_p), \quad (F, G) \longmapsto \bar{A}_m F + \bar{B}_m G.$$

Suppose

$$\bar{A}_m F + \bar{B}_m G = 0.$$

Since $\bar{A}_m$ and $\bar{B}_m$ are coprime, we have

$$\bar{A}_m \mid G, \quad \bar{B}_m \mid F.$$

Thus there is a unique

$$H \in V_{d-a_m-b_m}(\mathbf{F}_p)$$

such that

$$(F, G) = (-\bar{B}_m H, \bar{A}_m H).$$

Hence

$$\dim_{\mathbf{F}_p} \ker \bar{\Psi}_d = d - a_m - b_m + 1.$$

Since $d \geq a_m + b_m$, it follows that

$$\begin{aligned} \dim_{\mathbf{F}_p} \operatorname{im} \bar{\Psi}_d &= (d - a_m + 1) + (d - b_m + 1) - (d - a_m - b_m + 1) \\ &= d + 1 \\ &= \dim_{\mathbf{F}_p} V_d(\mathbf{F}_p). \end{aligned}$$

Therefore $\bar{\Psi}_d$ is surjective.

The cokernel of Ψ_d is a finitely generated $\mathbf{Z}_p$ -module, and its reduction modulo the maximal ideal (p) is the cokernel of $\bar{\Psi}_d$, which is zero. Nakayama's lemma therefore gives

$$\operatorname{coker} \Psi_d = 0.$$

Thus Ψ_d is surjective. □

3.5 Transfer map for modular symbols

We will now descend the multiplication map (3.1) to a transfer map on modular symbols modulo p^m .

Define the group homomorphism

$$\chi_m : R_m^\times \rightarrow R_m^\times, \quad u \mapsto u^{p^{m-1}}.$$

Lemma 3.4. *Let*

$$\gamma \in M_2(\mathbf{Z}), \quad p \nmid \det \gamma.$$

Then

$$A_m \mid \gamma = A_m, \quad B_m \mid \gamma = \chi_m(\det \gamma) B_m \quad \text{in } R_m[X, Y].$$

Proof. Modulo p , one has

$$\bar{\alpha}_p \mid \gamma = \bar{\alpha}_p, \quad \bar{\beta}_p \mid \gamma = \det(\gamma) \bar{\beta}_p.$$

These identities follow by a direct calculation. Raising them to the power p^{m-1} gives the stated congruences modulo p^m , by the binomial theorem. □

Proposition 3.5. *The coefficient multiplication map of (3.1) induces a homomorphism*

$$\Psi_d : \mathbb{M}_{d-a_m}(R_m) \oplus \mathbb{M}_{d-b_m}(R_m) \longrightarrow \mathbb{M}_d(R_m).$$

Moreover, if $p \nmid n$, then

$$T_n \circ \Psi_d = \Psi_d(T_n \oplus \chi_m(n) T_n). \tag{3.2}$$

If

$$d \geq a_m + b_m = p^{m-1}(p^2 + 1),$$

then Ψ_d is surjective.

Proof. For

$$F \in V_{d-a_m}(R_m), \quad G \in V_{d-b_m}(R_m),$$

Lemma 3.4 gives

$$\begin{aligned} \Psi_d(F, G) \mid \gamma &= (A_m F + B_m G) \mid \gamma \\ &= (A_m \mid \gamma)(F \mid \gamma) + (B_m \mid \gamma)(G \mid \gamma) \\ &= A_m(F \mid \gamma) + \chi_m(\det \gamma) B_m(G \mid \gamma) \\ &= \Psi_d(F \mid \gamma, \chi_m(\det \gamma)(G \mid \gamma)). \end{aligned} \tag{3.3}$$

Since

$$\det S = \det U = 1,$$

it follows that

$$\Psi_d(F, G) \mid S = \Psi_d(F \mid S, G \mid S)$$

and

$$\Psi_d(F, G) \mid U = \Psi_d(F \mid U, G \mid U).$$

Hence Ψ_d preserves the Manin relations, and therefore descends to

$$\Psi_d : \mathbb{M}_{d-a_m}(R_m) \oplus \mathbb{M}_{d-b_m}(R_m) \longrightarrow \mathbb{M}_d(R_m).$$

Now let $p \nmid n$. Every matrix $M \in \mathcal{H}_n$ has determinant n , so (3.3) gives

$$\begin{aligned} T_n \Psi_d(F, G) &= \sum_{M \in \mathcal{H}_n} \Psi_d(F, G) \mid M \\ &= \sum_{M \in \mathcal{H}_n} \Psi_d(F \mid M, \chi_m(n)(G \mid M)) \\ &= \Psi_d(T_n F, \chi_m(n) T_n G). \end{aligned}$$

This proves (3.2).

Finally, if

$$d \geq p^{m-1}(p^2 + 1),$$

then the coefficient multiplication map is surjective by Proposition 3.3. Hence the induced map on the Manin quotients is also surjective. $\square$

3.6 For odd p

In this subsection, suppose that p is odd. For $\epsilon \in \{+1, -1\}$, the idempotents

$$e^\epsilon := \frac{1 + \epsilon \iota}{2}$$

are defined over $\mathbf{Z}_p$ and R_m . We have

$$\begin{aligned} \mathbb{M}_d^\epsilon(R_m) &= e^\epsilon \mathbb{M}_d(R_m), \\ P_d^\epsilon &= \left(\mathbb{M}_d^{\text{tf}} \right)^\epsilon = e^\epsilon \mathbb{M}_d^{\text{tf}}, \\ \left(\mathbb{M}_d^{\text{tf}} / p^m \mathbb{M}_d^{\text{tf}} \right)^\epsilon &= e^\epsilon \left(\mathbb{M}_d^{\text{tf}} / p^m \mathbb{M}_d^{\text{tf}} \right) = P_d^\epsilon / p^m P_d^\epsilon. \end{aligned} \tag{3.4}$$

Moreover,

$$\mathbb{M}_d(R_m) = \mathbb{M}_d^+(R_m) \oplus \mathbb{M}_d^-(R_m), \tag{3.5}$$

with e^ϵ acting as the identity on $\mathbb{M}_d^\epsilon(R_m)$.

Since p^{m-1} is odd, Lemma 3.4 gives

$$A_m \mid \iota = A_m, \quad B_m \mid \iota = -B_m \quad (3.6)$$

in $R_m[X, Y]$. Thus multiplication by A_m preserves the ι -sign, whereas multiplication by B_m reverses it.

Proposition 3.6. *Suppose that p is odd. For each*

$$\epsilon \in \{+1, -1\},$$

there is a map

$$\Psi_d^\epsilon : \mathbb{M}_{d-a_m}^\epsilon(R_m) \oplus \mathbb{M}_{d-b_m}^{-\epsilon}(R_m) \rightarrow \mathbb{M}_d^\epsilon(R_m),$$

satisfying

$$T_n \circ \Psi_d^\epsilon = \Psi_d^\epsilon(T_n \oplus \chi_m(n)T_n) \quad ((n, p) = 1). \quad (3.7)$$

Moreover, if

$$d \geq p^{m-1}(p^2 + 1),$$

then Ψ_d^ϵ is surjective. In particular, for $\epsilon = +1$, composing with the quotient map to $P_d^+/p^m P_d^+$ gives a surjection

$$\Phi_d : \mathbb{M}_{d-a_m}^+(R_m) \oplus \mathbb{M}_{d-b_m}^-(R_m) \twoheadrightarrow P_d^+/p^m P_d^+. \quad (3.8)$$

Proof. For $F \in \mathbb{M}_{d-a_m}(R_m)$ and $G \in \mathbb{M}_{d-b_m}(R_m)$, we have, by (3.6),

$$e^\epsilon \Psi_d(F, G) = \Psi_d(e^\epsilon F, e^{-\epsilon} G).$$

Thus, defining Ψ_d^ϵ as the restriction of Ψ_d to $\mathbb{M}_{d-a_m}^\epsilon(R_m) \oplus \mathbb{M}_{d-b_m}^{-\epsilon}(R_m)$, we see that the image of Ψ_d^ϵ lands in $\mathbb{M}_d^\epsilon(R_m)$. By Proposition 3.5, it is twisted Hecke-equivariant, and surjective if $d \geq p^{m-1}(p^2 + 1)$.

Composing with the natural surjection

$$\mathbb{M}_d(R_m) \twoheadrightarrow \mathbb{M}_d^{\text{tf}}/p^m \mathbb{M}_d^{\text{tf}}$$

and applying e^+ gives (3.8). □

3.7 Propagation of Hecke operator relations

The transfer maps above allow relations on Hecke operators in lower coefficient degrees to be propagated to higher degrees.

By abuse of notation, we extend the character χ_m to an endomorphism of the polynomial algebra in the prime-to- p Hecke operators by

$$\chi_m(T_n) := \chi_m(n)T_n = n^{p^{m-1}}T_n \quad ((n, p) = 1).$$

Let F be a polynomial, with coefficients in R_m , in finitely many prime-to- p Hecke operators. For

$$\epsilon \in \{+1, -1\},$$

the twisted Hecke equivariance of Proposition 3.6 gives

$$F \circ \Psi_d^\epsilon = \Psi_d^\epsilon(F \oplus \chi_m(F)). \quad (3.9)$$

Corollary 3.7 (One-step propagation). *Suppose that p is odd and*

$$d \geq p^{m-1}(p^2 + 1).$$

Let $\epsilon \in \{+1, -1\}$. If

$$F\mathbb{M}_{d-a_m}^\epsilon(R_m) = 0 \quad \text{and} \quad \chi_m(F)\mathbb{M}_{d-b_m}^{-\epsilon}(R_m) = 0,$$

then

$$F\mathbb{M}_d^\epsilon(R_m) = 0.$$

In particular, for $\epsilon = +1$,

$$F\left(P_d^+/p^m P_d^+\right) = 0 \quad \text{and} \quad F\left(L_d^+(R_m)\right) = 0.$$

Proof. The first assertion follows from (3.9) and the surjectivity of Ψ_d^ϵ granted by Proposition 3.5. The final assertion follows by passing to the quotient

$$\mathbb{M}_d^+(R_m) \twoheadrightarrow P_d^+/p^m P_d^+$$

and then restricting to $L_d^+(R_m)$. □

Theorem 3.8 (Weight propagation). *Suppose that p is odd, and let r be an even integer with $0 \leq r < 2p^{m-1}$. Let F be a polynomial, with coefficients in R_m , in finitely many prime-to- p Hecke operators. For $q \geq 0$ put*

$$\epsilon_q := (-1)^q.$$

Suppose that, for every q ,

$$\chi_m^q(F)\mathbb{M}_{r+2jp^{m-1}}^{\epsilon_q}(R_m) = 0 \quad \left(\frac{p+1}{2} \leq j < \frac{p^2+1}{2} \right).$$

Then, for every $j \geq \frac{p+1}{2}$ and every q ,

$$\chi_m^q(F)\mathbb{M}_{r+2jp^{m-1}}^{\epsilon_q}(R_m) = 0.$$

Consequently, for every even $d \geq p^{m-1}(p+1)$ satisfying $d \equiv r \pmod{2p^{m-1}}$,

$$F\left(P_d^+/p^m P_d^+\right) = 0 \quad \text{and} \quad F\left(L_d^+/p^m L_d^+\right) = 0.$$

Proof. For ease of reading, let us introduce

$$u := \frac{p(p-1)}{2}, \quad v := \frac{p+1}{2}, \quad J := u + v = \frac{p^2+1}{2}.$$

Then

$$a_m = 2p^{m-1}u, \quad b_m = 2p^{m-1}v.$$

Write

$$d_j := r + 2p^{m-1}j.$$

Then the transfer map can be written as

$$\Psi_{d_j}^\epsilon : \mathbb{M}_{d_j-u}^\epsilon(R_m) \oplus \mathbb{M}_{d_j-v}^{-\epsilon}(R_m) \rightarrow \mathbb{M}_{d_j}^\epsilon(R_m).$$

In this notation, the surjectivity condition of Proposition 3.6 can be written as $j \geq J$.

We argue by strong induction on j . There is nothing to prove for $v \leq j < J$. Suppose that $j \geq J$. We quickly check that

$$j - u \geq J - u = v, \quad j - v \geq J - v = u \geq v.$$

By Proposition 3.6,

$$\Psi_{d_j}^{\epsilon_q} : \mathbb{M}_{d_j-u}^{\epsilon_q}(R_m) \oplus \mathbb{M}_{d_j-v}^{-\epsilon_q}(R_m) \twoheadrightarrow \mathbb{M}_{d_j}^{\epsilon_q}(R_m)$$

is surjective. Recall that the second summand in the domain carries the twist χ_m .

By the induction hypothesis,

$$\chi_m^q(F) \mathbb{M}_{d_j-u}^{\epsilon_q}(R_m) = 0$$

and

$$\chi_m^{q+1}(F) \mathbb{M}_{d_j-v}^{\epsilon_q+1}(R_m) = 0.$$

By Corollary 3.7,

$$\chi_m^q(F) \mathbb{M}_{d_j}^{\epsilon_q}(R_m) = 0,$$

finishing the induction.

Taking $q = 0$ gives the assertion on $\mathbb{M}_d^+(R_m)$, and hence on $P_d^+/p^m P_d^+$ and $L_d^+/p^m L_d^+$. $\square$

Remark 3.9. More generally, the Hecke relation may depend periodically on the coefficient degree. The same induction applies to a finite family F_d , provided that the family is closed under the two transitions

$$(d, F_d, \epsilon) \mapsto (d - a_m, F_d, \epsilon)$$

and

$$(d, F_d, \epsilon) \mapsto (d - b_m, \chi_m(F_d), -\epsilon).$$

For $p = 2$, the determinant twist is trivial. We state the simpler propagation statement.

Theorem 3.10 (Weight propagation for $p = 2$). *Suppose that $p = 2$ and $m \geq 2$. Let r be an even integer with*

$$0 \leq r < 2^{m-1},$$

and let F be a polynomial, with coefficients in R_m , in finitely many odd Hecke operators. Suppose that

$$F \mathbb{M}_{r+j2^{m-1}}(R_m) = 0 \quad (2 \leq j \leq 4).$$

Then

$$F \mathbb{M}_{r+j2^{m-1}}(R_m) = 0$$

for every $j \geq 2$.

Consequently, for every even $d \geq 2^m$ satisfying $d \equiv r \pmod{2^{m-1}}$,

$$F \left(P_d^+ / 2^m P_d^+ \right) = 0 \quad \text{and} \quad F \left(L_d^+ / 2^m L_d^+ \right) = 0.$$

Proof. By strong induction on j , using the degree shifts

$$a_m = 2^m, \quad b_m = 3 \cdot 2^{m-1}.$$

$\square$

3.8 Calculus of linear relations

To obtain higher congruences, we need polynomial relations in divided Hecke operators such as T/p^α . Division need not define an endomorphism on the finite modular-symbol modules used in the transfer argument. We therefore encode it by the linear relation $Tx = p^\alpha y$. Such relations pass through the transfer maps and yield ordinary operator congruences after passage to the free lattice.

In this section, we will make this precise. Let R be a commutative ring. If M is an R -module, we define an R -linear relation on M to be an R -submodule of $M \oplus M$. We denote by $\text{LinRel}_R(M)$ the set of all R -linear relations on M . Any R -linear map $T \in \text{End}_R(M)$ gives rise to such a relation via its graph

$$\Gamma_T := \{(x, Tx) : x \in M\}.$$

If $\mathcal{A} \in \text{LinRel}_R(M)$ and $x, y \in M$, write $x\mathcal{A}y$ to mean that $(x, y) \in \mathcal{A}$. For a sequence $x_0, x_1, \dots, x_n \in M$, write

$$x_0\mathcal{A}x_1\mathcal{A}\dots\mathcal{A}x_n$$

to mean that

$$x_0\mathcal{A}x_1, \quad x_1\mathcal{A}x_2, \quad \dots \quad x_{n-1}\mathcal{A}x_n.$$

Let

$$\mathcal{A}(x) := \{y \in M : (x, y) \in \mathcal{A}\}.$$

Define the domain of $\mathcal{A}$ as

$$\text{dom } \mathcal{A} := \{x \in M : \mathcal{A}(x) \neq \emptyset\}.$$

If $\mathcal{A}, \mathcal{B} \in \text{LinRel}_R(M)$ and $r \in R$, define

$$\mathcal{A} + \mathcal{B} := \{(x, y + z) : (x, y) \in \mathcal{A}, (x, z) \in \mathcal{B}\},$$

$$r\mathcal{A} = \{(x, ry) : (x, y) \in \mathcal{A}\},$$

$$\mathcal{A}\mathcal{B} = \{(x, z) : \exists y \in M, (x, y) \in \mathcal{B}, (y, z) \in \mathcal{A}\}.$$

If $f : M \rightarrow N$ is R -linear, we get a map $f_* : \text{LinRel}_R(M) \rightarrow \text{LinRel}_R(N)$ defined as follows: we have an R -linear map

$$f \oplus f : M \oplus M \rightarrow N \oplus N,$$

and for every $\mathcal{A} \in \text{LinRel}_R(M)$, we can define

$$f_*(\mathcal{A}) := (f \oplus f)(\mathcal{A}).$$

Now we make a more universal definition. Define a *presentation of a linear relation over R* to be a triple $\mathfrak{P} := (P, p_0, p_1)$ of an R -module P with two distinguished elements $p_0, p_1 \in P$. If $\mathfrak{P}$ is a presentation of a linear relation over R and M is an R -module, let

$$\mathfrak{P}(M) := \{(\phi(p_0), \phi(p_1)) : \phi \in \text{Hom}_R(P, M)\}.$$

Thus $\mathfrak{P}(M) \in \text{LinRel}_R(M)$. Moreover, if $f : M \rightarrow N$ is R -linear, then

$$f_*\mathfrak{P}(M) \subset \mathfrak{P}(N).$$

Consequently,

$$f(\operatorname{dom} \mathfrak{P}(M)) \subseteq \operatorname{dom} \mathfrak{P}(N).$$

In particular, if f is surjective and $\operatorname{dom} \mathfrak{P}(M) = M$, then $\operatorname{dom} \mathfrak{P}(N) = N$. Presentations also commute with finite direct sums:

$$\mathfrak{P}(M_1 \oplus M_2) = \mathfrak{P}(M_1) \oplus \mathfrak{P}(M_2),$$

under the natural identification

$$(M_1 \oplus M_2) \oplus (M_1 \oplus M_2) \simeq (M_1 \oplus M_1) \oplus (M_2 \oplus M_2).$$

This follows from

$$\operatorname{Hom}_R(P, M_1 \oplus M_2) \simeq \operatorname{Hom}_R(P, M_1) \oplus \operatorname{Hom}_R(P, M_2).$$

In particular, full domain on both summands implies full domain on their direct sum.

We now define the sum, product, and scalar action on presentations of linear relations over R . Let $\mathfrak{P} = (P, p_0, p_1)$ and $\mathfrak{Q} = (Q, q_0, q_1)$ be two such relations. Define

$$\mathfrak{P} + \mathfrak{Q} := \left(\frac{P \oplus Q}{R(p_0, -q_0)}, \bar{p}_0 = \bar{q}_0, \overline{p_1 + q_1} \right)$$

where bars denote images in the quotient. Also define

$$\mathfrak{P}\mathfrak{Q} := \left(\frac{Q \oplus P}{R(q_1, -p_0)}, \bar{q}_0, \bar{p}_1 \right).$$

For $r \in R$, define $r\mathfrak{P} := (P, p_0, rp_1)$. Then, for every M ,

$$(\mathfrak{P} + \mathfrak{Q})(M) = \mathfrak{P}(M) + \mathfrak{Q}(M),$$

$$(\mathfrak{P}\mathfrak{Q})(M) = \mathfrak{P}(M)\mathfrak{Q}(M),$$

$$(r\mathfrak{P})(M) = r\mathfrak{P}(M).$$

Finally, let

$$\mathbf{1}_R := (R, 1, 1).$$

Then there are canonical isomorphisms

$$\mathfrak{P}\mathbf{1}_R \cong \mathbf{1}_R\mathfrak{P} \cong \mathfrak{P}.$$

Let us remark that we can make these definitions functorially. Consider the functor

$$\mathcal{U} : R\text{-Mod} \rightarrow R\text{-Mod}, \quad \mathcal{U}(M) := M \oplus M.$$

Then a *linear relation* $\mathcal{P}$ over R is a subfunctor $\mathcal{P} \subset \mathcal{U}$. If $\mathfrak{P} = (P, p_0, p_1)$ is a presentation of a linear relation over R , then the image subfunctor

$$\mathcal{P}_{\mathfrak{P}} := \operatorname{im}(\operatorname{Hom}_R(P, -) \rightarrow \mathcal{U})$$

is a linear relation over R . Note however that not every linear relation will have a presentation.

We now specify how to evaluate a polynomial at linear relations. For a linear relation $\mathcal{A}$ on M , put

$$\mathcal{A}^0 := \Gamma_{\operatorname{id}_M}, \quad \mathcal{A}^{n+1} := \mathcal{A}\mathcal{A}^n.$$

Thus, for $n \geq 1$,

$$\mathcal{A}^n(x_0) = \{x_n \in M : \exists x_1, \dots, x_{n-1} \in M, x_0 \mathcal{A} x_1 \mathcal{A} \cdots \mathcal{A} x_n\}.$$

For a nonzero polynomial

$$F(X) = \sum_{j=0}^n c_j X^j \in R[X],$$

define

$$F(\mathcal{A}) := \sum_{j:c_j \neq 0} c_j \mathcal{A}^j.$$

For the zero polynomial, put $F(\mathcal{A}) := \Gamma_0$. More generally, if

$$F(X_1, \dots, X_s) = \sum_j c_j X_1^{j_1} \cdots X_s^{j_s},$$

we define

$$F(\mathcal{A}_1, \dots, \mathcal{A}_s) := \sum_{j:c_j \neq 0} c_j \mathcal{A}_1^{j_1} \cdots \mathcal{A}_s^{j_s}.$$

The order of composition is the displayed order, with the rightmost relation applied first. Different monomials may use different intermediate elements.

These definitions use the expanded polynomial. Evaluation at multivalued relations need not respect polynomial addition or multiplication; in particular, algebraic simplifications of expressions in relations are not automatically valid. For commuting endomorphisms $T_1, \dots, T_s$, however,

$$F(\Gamma_{T_1}, \dots, \Gamma_{T_s}) = \Gamma_{F(T_1, \dots, T_s)}.$$

For $F(X) = \sum_{j=0}^n c_j X^j$, any common chain

$$x_0 \mathcal{A} x_1 \mathcal{A} \cdots \mathcal{A} x_n$$

gives

$$\left(x_0, \sum_{j=0}^n c_j x_j \right) \in F(\mathcal{A}).$$

Thus a computation using a common chain supplies an element of the polynomial relation defined above.

We will now define the *divided operator* linear relation. Let $T, S \in R$. Define a presentation of a linear relation over R by

$$\mathfrak{D}_{T,S} := \left(\frac{R \oplus R}{R(Te_0 - Se_1)}, \bar{e}_0, \bar{e}_1 \right), \quad e_0 := (1, 0), \quad e_1 := (0, 1).$$

Then on an R -module M ,

$$\mathfrak{D}_{T,S}(M) = \{(x, y) : Tx = Sy\}.$$

If $\text{dom } \mathfrak{D}_{T,S}(M) = M$, we will say that M admits a *division relation of T by S* and symbolically write

$${}''\frac{T}{S} := \mathfrak{D}_{T,S}(M).$$

We now specialize a bit more towards our intended applications. Let A be a commutative $\mathbf{Z}_p$ -algebra, and let L be an A -module that is finite free over $\mathbf{Z}_p$. Fix $m \geq 1$, and let M be an $A/p^m A$ -module equipped with an $A/p^m A$ -linear surjection

$$q : M \twoheadrightarrow L/p^m L.$$

We apply the preceding definitions over the ring $A/p^m A$. Elements of A acting on M are understood through their images in $A/p^m A$.

Let $\mathcal{B}$ be a linear relation on M , and let $0 \leq b \leq m$. We write

$$\mathcal{B} \equiv 0 \pmod{p^b} \quad \text{on } M$$

if

$$\mathcal{B}(x) \cap p^b M \neq \emptyset \quad \text{for every } x \in M.$$

Equivalently, for every $x \in M$, there exists $y \in p^b M$ such that $(x, y) \in \mathcal{B}$.

For an endomorphism T of M , this definition gives

$$\Gamma_T \equiv 0 \pmod{p^b} \iff TM \subseteq p^b M.$$

More generally, for $T \in A$,

$$\mathcal{B}\Gamma_T \equiv 0 \pmod{p^b}$$

means that

$$\mathcal{B}(Tu) \cap p^b M \neq \emptyset \quad \text{for every } u \in M.$$

For a presentation $\mathfrak{P}$ over $A/p^m A$, put

$$\mathfrak{I} := \mathfrak{D}_{1, p^b} \mathfrak{P}.$$

Then

$$\mathfrak{I}(M) = \{(x, \rho) : \exists y \in \mathfrak{P}(M)(x), y = p^b \rho\}.$$

Consequently,

$$\mathfrak{P}(M) \equiv 0 \pmod{p^b} \iff \text{dom } \mathfrak{I}(M) = M.$$

Thus congruence conditions for presented relations pass through surjective module homomorphisms and hold on a finite direct sum whenever they hold on each summand.

Proposition 3.11 (Division and retained precision). *In the preceding setup, let $T_1, \dots, T_s \in A$, and let $1 \leq \alpha_i < m$. Put*

$$\mathcal{Z}_i := \mathfrak{D}_{T_i, p^{\alpha_i}}(M).$$

For each i , if $\text{dom } \mathcal{Z}_i = M$, then

$$T_i L \subseteq p^{\alpha_i} L.$$

Consequently,

$$Z_i := \frac{T_i}{p^{\alpha_i}}$$

defines an endomorphism of L .

Suppose now that all the Z_i preserve L , whether established this way or separately, and put

$$h := m - \max_i \alpha_i.$$

Let $F \in \mathbf{Z}_p[X_1, \dots, X_s]$, let $S \in A$, and let $0 \leq b \leq h$. If

$$F(Z_1, \dots, Z_s)\Gamma_S \equiv 0 \pmod{p^b} \quad \text{on } M,$$

then

$$SF(Z_1, \dots, Z_s)L \subseteq p^b L.$$

Here the coefficients of F on the source are understood through their images in $A/p^m A$.

Proof. Suppose first that $\text{dom } \mathcal{Z}_i = M$. For $u \in L$, choose $x \in M$ with $q(x) = u \pmod{p^m L}$, and choose $y \in M$ satisfying

$$T_i x = p^{\alpha_i} y.$$

Applying q gives

$$T_i u \in p^{\alpha_i} L + p^m L = p^{\alpha_i} L.$$

This proves the first assertion.

Now assume that all the Z_i preserve L . Let

$$q_h : M \twoheadrightarrow L/p^h L$$

be the composite of q with reduction modulo p^h . If $(x, y) \in \mathcal{Z}_i$, applying q and cancelling p^{α_i} on the free lattice gives

$$q_h(y) = Z_i q_h(x),$$

since $h \leq m - \alpha_i$. Thus every choice of output of $\mathcal{Z}_i$ has, after applying q_h , the same value as the integral operator Z_i . Following the intermediate elements in each monomial, we therefore obtain

$$q_h(y) = F(Z_1, \dots, Z_s)q_h(x)$$

whenever

$$(x, y) \in F(\mathcal{Z}_1, \dots, \mathcal{Z}_s).$$

For any $u \in L$, choose $x \in M$ lifting its class modulo $p^m L$. By hypothesis, there is $y \in p^b M$ such that

$$(Sx, y) \in F(\mathcal{Z}_1, \dots, \mathcal{Z}_s).$$

Consequently,

$$F(Z_1, \dots, Z_s)Su \in p^b L + p^h L = p^b L.$$

Since A is commutative, $S, Z_1, \dots, Z_s$ commute on $L \otimes_{\mathbf{Z}_p} \mathbf{Q}_p$, and hence on L . This gives the stated inclusion. $\square$

3.9 Propagation of divided Hecke relations

Fix $m \geq 1$, and assume $m \geq 2$ if $p = 2$. Recall that $R_m = \mathbf{Z}/p^m \mathbf{Z}$, and put

$$\mathbb{A} := \mathbf{Z}_p[t_n : (n, p) = 1], \quad \mathbb{A}_m := \mathbb{A}/p^m \mathbb{A} = R_m[t_n : (n, p) = 1].$$

For odd p , put $\epsilon_q = (-1)^q$. We regard

$$\mathbb{M}_d^{\epsilon_q}(R_m)$$

as an $\mathbb{A}_m$ -module by letting t_n act as $\chi_m(n)^q T_n$. With these actions, the transfer map

$$\Psi_d^{\epsilon_q} : \mathbb{M}_{d-a_m}^{\epsilon_q}(R_m) \oplus \mathbb{M}_{d-b_m}^{\epsilon_q+1}(R_m) \longrightarrow \mathbb{M}_d^{\epsilon_q}(R_m)$$

is $\mathbb{A}_m$ -linear and is surjective when $d \geq a_m + b_m$. Here the two source modules carry the actions associated with orientations q and $q + 1$, respectively.

For $p = 2$, let t_n act as T_n on $\mathbb{M}_d(R_m)$. The transfer map

$$\Psi_d : \mathbb{M}_{d-a_m}(R_m) \oplus \mathbb{M}_{d-b_m}(R_m) \longrightarrow \mathbb{M}_d(R_m)$$

is $\mathbb{A}_m$ -linear and surjective in the same range.

Proposition 3.12 (Propagation of presented relations). *Let $\mathfrak{P}$ be a presentation of a linear relation over $\mathbb{A}_m$.*

Suppose first that p is odd, and let r be an even integer with $0 \leq r < 2p^{m-1}$. Assume that

$$\text{dom } \mathfrak{P}(\mathbb{M}_{r+2jp^{m-1}}^{\epsilon_q}(R_m)) = \mathbb{M}_{r+2jp^{m-1}}^{\epsilon_q}(R_m)$$

for every $q \geq 0$ and every integer j satisfying

$$\frac{p+1}{2} \leq j < \frac{p^2+1}{2},$$

with the $\mathbb{A}_m$ -actions specified above. Then the same equality holds for every $j \geq (p+1)/2$ and every $q \geq 0$.

For $p = 2$, let r be an even integer with $0 \leq r < 2^{m-1}$. If

$$\text{dom } \mathfrak{P}(\mathbb{M}_{r+j2^{m-1}}(R_m)) = \mathbb{M}_{r+j2^{m-1}}(R_m) \quad (2 \leq j \leq 4),$$

then the same equality holds for every $j \geq 2$.

In both cases, the same conclusion holds with the full-domain condition replaced by the congruence

$$\mathfrak{P}(\mathbb{M}_{r+2jp^{m-1}}^{\epsilon_q}(R_m)) \equiv 0 \pmod{p^b},$$

or its unsplit analogue for $p = 2$, where $0 \leq b \leq m$.

Proof. Use the same strong induction on j as in Theorems 3.8 and 3.10. At each step, the induction hypothesis gives full domain on both source modules, hence on their direct sum. Full domain then passes through the surjective transfer map.

For congruences, apply the same argument to the presentation

$$\mathfrak{D}_{1,p^b} \mathfrak{P}.$$

□

As r varies, the initial degrees are exactly the even degrees in the ranges

$$b_m \leq d < a_m + b_m \quad (p \text{ odd}),$$

and

$$2^m \leq d < 5 \cdot 2^{m-1} \quad (p = 2).$$

For odd p , the specified module structures are periodic in q with period $p - 1$, so only finitely many orientations need to be checked.

As in Remark 3.9, the presentation may depend on degree residues and orientation. The same induction applies provided that, at each transfer step, the source conditions are those of the target presentation evaluated on the two source modules with their respective $\mathbb{A}_m$ -actions.

Thus any family closed under the two degree transitions propagates in the same way. One may likewise restrict to degree residues closed under these transitions.

To obtain operator congruences, take orientation $q = 0$ and use

$$\mathbb{M}_d^+(R_m) \twoheadrightarrow P_d^+/p^m P_d^+ \quad (p \text{ odd}),$$

or

$$\mathbb{M}_d(R_m) \twoheadrightarrow \mathbb{M}_d^{\text{tf}}/2^m \mathbb{M}_d^{\text{tf}} \quad (p = 2).$$

Give the free lattices their $\mathbb{A}$ -module structures by letting t_n act as T_n . Proposition 3.11, applied with $A = \mathbb{A}$, then establishes integrality of the divided operators and the resulting polynomial congruences on the free lattice, under its stated hypotheses.

These conclusions restrict to L_d^+ . Indeed, the numerator Hecke operators preserve L_d^+ , and saturation implies that their integral divided operators preserve it as well. For $p = 2$, these operators also commute with ι , by cancellation on the torsion-free module, and hence preserve P_d^+ . Finally, saturation gives

$$L_d^+ \cap p^b P_d^+ = p^b L_d^+$$

for odd p , and

$$L_d^+ \cap 2^b \mathbb{M}_d^{\text{tf}} = 2^b L_d^+$$

for $p = 2$.

Relations with prescribed inputs are included in the same argument. For example,

$$F(\mathcal{Z}_1, \dots, \mathcal{Z}_s) \Gamma_S \equiv 0 \pmod{p^b}$$

is a presented relation, with the action of S twisted along with the numerator operators.

3.10 Recursive computation of Manin quotients and ideal images

The transfer map allows a recursive and fast computation of the Manin quotients. We describe the procedure here. Fix $m \geq 1$, assuming $m \geq 2$ if $p = 2$. Let $d \geq 0$ be even and $R_m = \mathbf{Z}/p^m \mathbf{Z}$. We use the convention that $V_d(R_m) = 0$ if $d < 0$.

Suppose $d < a_m + b_m$. By the proof of Proposition 3.3, the transfer map

$$\Psi_d : V_{d-a_m}(R_m) \oplus V_{d-b_m}(R_m) \longrightarrow V_d(R_m)$$

is injective. In fact, its reduction over $\mathbf{F}_p$ is injective, hence it is split injective. Thus there exists a free R_m -submodule $W_d \subseteq V_d(R_m)$ such that

$$V_d(R_m) = A_m V_{d-a_m}(R_m) \oplus B_m V_{d-b_m}(R_m) \oplus W_d. \quad (3.10)$$

The complement W_d need not be stable under $\text{SL}_2(\mathbf{Z})$. Since W_d is a direct summand of a finite free module over the local ring R_m , it is itself finite free.

Put

$$H_d := \mathbb{M}_{d-a_m}(R_m) \oplus \mathbb{M}_{d-b_m}(R_m).$$

Define an R_m -linear surjection

$$\rho_d : V_d(R_m) \longrightarrow H_d \oplus W_d, \quad A_m F + B_m G + w \longmapsto ([F], [G], w).$$

Let $\mathcal{R}_d$ be the R_m -submodule generated by

$$\rho_d(w + w \mid S), \quad \rho_d(w + w \mid U + w \mid U^2),$$

as w ranges over a basis of W_d .

Proposition 3.13. *The map*

$$([F], [G], w) \mapsto [A_m F + B_m G + w]$$

induces an isomorphism of R_m -modules

$$(H_d \oplus W_d)/\mathcal{R}_d \simeq \mathbb{M}_d(R_m).$$

Proof. For each nonnegative degree e , put

$$N_e := V_e(R_m)(1 + S) + V_e(R_m)(1 + U + U^2).$$

Terms involving negative degrees below are omitted. By (3.10) and the definition of ρ_d , we have

$$\ker \rho_d = A_m N_{d-a_m} + B_m N_{d-b_m}.$$

Since A_m and B_m are invariant under S and U ,

$$N_d = A_m N_{d-a_m} + B_m N_{d-b_m} + W_d(1 + S) + W_d(1 + U + U^2).$$

Consequently,

$$\ker \rho_d \subseteq N_d, \quad \rho_d(N_d) = \mathcal{R}_d.$$

Therefore

$$\mathbb{M}_d(R_m) = \frac{V_d(R_m)}{N_d} \simeq \frac{V_d(R_m)/\ker \rho_d}{N_d/\ker \rho_d} \simeq \frac{H_d \oplus W_d}{\mathcal{R}_d}.$$

□

The Hecke actions can also be computed recursively. For $(n, p) = 1$, define an endomorphism of $H_d \oplus W_d$ by

$$\tilde{T}_n(x, y, w) = (T_n x, \chi_m(n) T_n y, 0) + \rho_d \left(\sum_{\gamma \in \mathcal{H}_n} w \mid \gamma \right).$$

By twisted Hecke equivariance of the transfer map, this endomorphism preserves $\mathcal{R}_d$ and induces T_n on the quotient of Proposition 3.13. Thus only the Hecke images of a basis of W_d need to be computed directly.

For odd p , the construction is compatible with the plus/minus decomposition. Applying the projections $e^\pm$, we may choose complements separately in the two eigenspaces:

$$V_d(R_m)^\epsilon = A_m V_{d-a_m}(R_m)^\epsilon \oplus B_m V_{d-b_m}(R_m)^{-\epsilon} \oplus W_d^\epsilon.$$

Taking $W_d = W_d^+ \oplus W_d^-$ gives an ι -stable complement. Since $A_m \mid \iota = A_m$ and $B_m \mid \iota = -B_m$, the map ρ_d is equivariant for the involution

$$(x, y, w) \mapsto (\iota x, -\iota y, \iota w)$$

on $H_d \oplus W_d$. In particular, $\mathcal{R}_d$ is stable under this involution. Applying $e^\epsilon = (1 + \epsilon)/2$ therefore gives

$$\mathbb{M}_d^\epsilon(R_m) \simeq \frac{\mathbb{M}_{d-a_m}^\epsilon(R_m) \oplus \mathbb{M}_{d-b_m}^{-\epsilon}(R_m) \oplus W_d^\epsilon}{e^\epsilon \mathcal{R}_d}.$$

To compute $e^\epsilon \mathcal{R}_d$, form the relations from a basis of all of W_d and then project. In particular, the U -relations must include starting vectors of both signs. The induced Hecke operators preserve these signed quotients. For $p = 2$, we use the unsplit construction.

The same construction permits recursive verification of presented linear relations. Give the first lower quotient the usual $\mathbb{A}_m$ -action and the second the action $t_n \mapsto \chi_m(n)T_n$. The maps induced by multiplication by A_m and B_m are then $\mathbb{A}_m$ -linear. Let $\mathfrak{P}$ be a presentation over $\mathbb{A}_m$. If $\mathfrak{P}$ has full domain on both lower quotients, functoriality shows that their images lie in

$$\text{dom } \mathfrak{P}(\mathbb{M}_d(R_m)).$$

It therefore suffices to check

$$[w] \in \text{dom } \mathfrak{P}(\mathbb{M}_d(R_m))$$

for each vector w in a basis of W_d . Indeed, this domain is a submodule, and these classes together with the transferred generators generate $\mathbb{M}_d(R_m)$. Likewise, a congruence

$$\mathfrak{P}(\mathbb{M}_d(R_m)) \equiv 0 \pmod{p^b}$$

can be verified recursively by applying this argument to the presentation $\mathfrak{D}_{1,p^b}\mathfrak{P}$.

The preceding constructions also apply to ideal images. Let $I \subseteq \mathbb{A}_m$ be an ideal. For odd p , give $\mathbb{M}_d^\epsilon(R_m)$ and $\mathbb{M}_{d-a_m}^\epsilon(R_m)$ the action $t_n \mapsto \chi_m(n)^q T_n$, and $\mathbb{M}_{d-b_m}^{-\epsilon}$ the action $t_n \mapsto \chi_m(n)^{q+1} T_n$. The transfer map restricts to

$$\Psi_d^\epsilon : \mathbb{M}_{d-a_m}^\epsilon(R_m) \oplus \mathbb{M}_{d-b_m}^{-\epsilon}(R_m) \longrightarrow \mathbb{M}_d^\epsilon(R_m).$$

This restriction is surjective when $d \geq a_m + b_m$. Indeed, the transfer is $\mathbb{A}_m$ -linear with the specified actions, and a surjective $\mathbb{A}_m$ -linear map $f : M \rightarrow N$ satisfies $f(IM) = IN$. For $p = 2$, the same statement holds for the unsplit modules, with no twist.

Since taking ideal images also commutes with finite direct sums, the same induction proves propagation of ordinary operator relations and full-domain conditions for presented relations on these ideal images, with the initial degree ranges of Proposition 3.12. Families depending on degree and orientation must satisfy the same transition compatibility conditions. Congruences of presented relations are interpreted within the ideal images, so the terminal submodule is $p^b \mathbb{M}_d^\epsilon(R_m)$.

For odd p and $d < a_m + b_m$, the recursive presentation gives

$$\begin{aligned} \mathbb{M}_d^\epsilon(R_m) &= \Psi_d^\epsilon \left(\mathbb{M}_{d-a_m}^\epsilon(R_m) \oplus \mathbb{M}_{d-b_m}^{-\epsilon}(R_m) \right) \\ &\quad + \sum_{f \in I} f[W_d^\epsilon], \end{aligned}$$

where $[W_d^\epsilon]$ denotes the image of W_d^ϵ in the Manin quotient, and f acts through the target orientation. If $I = (f_1, \dots, f_s)$, this equality becomes

$$\begin{aligned} \mathbb{M}_d^\epsilon(R_m) &= \Psi_d^\epsilon \left(\mathbb{M}_{d-a_m}^\epsilon(R_m) \oplus \mathbb{M}_{d-b_m}^{-\epsilon}(R_m) \right) \\ &\quad + \sum_{j=1}^s f_j[W_d^\epsilon]. \end{aligned}$$

For $p = 2$, use the corresponding unsplit construction.

For recursive verification of a presented relation $\mathfrak{P}$, suppose it has full domain on both lower ideal images. It then suffices to check

$$f_j[w] \in \text{dom } \mathfrak{P}(\mathbb{M}_d^\epsilon(R_m))$$

for each ideal generator f_j and each vector w in a basis of W_d^ϵ . Congruences are treated by applying the same argument to $\mathfrak{D}_{1,p^b}\mathfrak{P}$.

Part II

Computations and classifications

Throughout this part, let f be a normalized cuspidal level-one eigenform of weight k , and let $\mathcal{O}_f$ be the ring of integers of a finite extension of $\mathbf{Q}_p$ containing the coefficients of f . We put

$$d := k - 2, \quad r \equiv d \pmod{p^{m-1}(p-1)}, \quad 0 \leq r < p^{m-1}(p-1).$$

4 The prime 2: classification modulo 256

For every even residue $r \bmod 128$, put

$$c_{3,r} := 1 + 3^{r+1} \pmod{256}, \quad c_{5,r} := 1 + 5^{r+1} \pmod{256},$$

and define

$$X_r := T_3 - c_{3,r}.$$

Also put

$$\varepsilon_r := \begin{cases} 1, & r \equiv 0, 6, 8, 14 \pmod{16}, \\ 0, & r \equiv 2, 4, 10, 12 \pmod{16}. \end{cases}$$

Although r is taken modulo 128, these quantities depend only on $r \bmod 64$.

4.1 Certified Hecke operator identities

Put

$$F_r(Z) := Z^2 - \varepsilon_r Z.$$

The computation is performed at precision 256. For $m = 8$, the Dickson multipliers have degrees

$$a_8 = 256, \quad b_8 = 384, \quad \text{with } a_8 + b_8 = 640.$$

Hence the exact induction base in Proposition 3.12 consists, for every even $0 \leq r < 128$, of the three degrees

$$d = r + j \cdot 128, \quad 2 \leq j \leq 4.$$

Equivalently, these are the 192 even degrees

$$256 \leq d \leq 638.$$

For every such degree d , put

$$\mathcal{Z}_r := {}'' \frac{t_3 - c_{3,r}}{2^7} {}'' = \mathfrak{D}_{t_3 - c_{3,r}, 2^7}(\mathbb{M}_d(\mathbf{Z}/256\mathbf{Z})),$$

where $r \equiv d \pmod{128}$ and t_3 acts as T_3 . The computation [Rus26, `classification_mod_256.ipynb`] verifies

$$\text{dom } \mathcal{Z}_r = \mathbb{M}_d(\mathbf{Z}/256\mathbf{Z}), \quad F_r(\mathcal{Z}_r) \equiv 0 \pmod{2},$$

and also verifies the ordinary relation

$$(T_5 - c_{5,r})\mathbb{M}_d(\mathbf{Z}/256\mathbf{Z}) = 0.$$

Since both a_8 and b_8 are divisible by 128, the quantities $c_{3,r}$, $c_{5,r}$, and ε_r are unchanged under the two transitions of Remark 3.9. Thus these families satisfy the compatibility required for propagation.

Proposition 3.12 propagates these conditions to every even degree

$$d \geq 256.$$

The remaining even degrees

$$0 \leq d < 256$$

are verified directly. By Proposition 3.11, with $m = 8$, $\alpha_1 = 7$, $b = 1$, and $S = 1$, for

$$r \equiv d \pmod{128},$$

the operator

$$Z_r := \frac{T_3 - c_{3,r}}{2^7}$$

belongs to

$$\text{End}_{\mathbf{Z}_2}(\mathbb{M}_d^{\text{tf}}),$$

preserves P_d^+ and L_d^+ , and satisfies

$$(Z_r^2 - \varepsilon_r Z_r) \mathbb{M}_d^{\text{tf}} \subseteq 2\mathbb{M}_d^{\text{tf}}, \quad (Z_r^2 - \varepsilon_r Z_r) L_d^+ \subseteq 2L_d^+. \quad (4.1)$$

Likewise, Theorem 3.10, together with the direct verification in the lower degrees, gives

$$(T_5 - c_{5,r}) L_d^+ \subseteq 256 L_d^+. \quad (4.2)$$

Thus both relations hold in every even coefficient degree d .

4.2 Congruences for a_3 and a_5

By the Eichler–Shimura comparison recalled in Section 3.3, choose a primitive simultaneous eigenvector

$$v_f \in L_d^+ \otimes_{\mathbf{Z}_2} \mathcal{O}_f$$

for the prime-to-2 Hecke eigensystem of f .

Since Z_r preserves L_d^+ , its eigenvalue on v_f is integral. Thus

$$z_3 := \frac{a_3(f) - c_{3,r}}{2^7} \in \mathcal{O}_f.$$

Evaluating (4.1) on v_f gives

$$z_3^2 - \varepsilon_r z_3 \in 2\mathcal{O}_f.$$

If $\varepsilon_r = 0$, then

$$z_3^2 \in 2\mathcal{O}_f,$$

so $v_2(z_3) > 0$. Hence

$$v_2(a_3(f) - c_{3,r}) > 7,$$

which is precisely

$$a_3(f) \equiv_{\text{val}} c_{3,r} \pmod{256}.$$

If $\varepsilon_r = 1$, then

$$z_3(z_3 - 1) \in 2\mathcal{O}_f.$$

The two factors differ by a unit, so exactly one of them belongs to the maximal ideal of $\mathcal{O}_f$. Therefore

$$z_3 \equiv 0 \quad \text{or} \quad 1 \pmod{\mathfrak{m}_{\mathcal{O}_f}},$$

and consequently

$$a_3(f) \equiv_{\text{val}} c_{3,r} \quad \text{or} \quad c_{3,r} + 2^7 \pmod{256}.$$

Finally, evaluating (4.2) on v_f gives the literal, and hence valuative, congruence

$$a_5(f) \equiv_{\text{lit}} c_{5,r} \pmod{256}.$$

Thus

$$a_3(f) \equiv_{\text{val}} \begin{cases} c_{3,r} \pmod{256}, & \varepsilon_r = 0, \\ c_{3,r} \text{ or } c_{3,r} + 128 \pmod{256}, & \varepsilon_r = 1, \end{cases} \quad a_5(f) \equiv_{\text{lit}} c_{5,r} \pmod{256}. \quad (4.3)$$

4.3 Packet selection

Consider the two cyclotomic systems $\text{Cyc}_{0,k-1}$ and $\text{Cyc}_{16,k-17}$. At an odd prime ℓ , their values are

$$\text{Cyc}_{0,k-1}(\ell) = 1 + \ell^{k-1}$$

and

$$\text{Cyc}_{16,k-17}(\ell) = \ell^{16} + \ell^{k-17},$$

where the exponents are understood modulo 64. In both cases, the two exponents have sum $k - 1$. Since every odd unit modulo 256 has exponent dividing 64, these systems depend only on the coefficient-degree residue $d = k - 2 \pmod{64}$.

For $r \equiv d \pmod{128}$, a direct calculation gives

$$\begin{aligned} \text{Cyc}_{0,k-1}(3) &\equiv c_{3,r}, & \text{Cyc}_{0,k-1}(5) &\equiv c_{5,r}, \\ \text{Cyc}_{16,k-17}(3) &\equiv c_{3,r} + 128, & \text{Cyc}_{16,k-17}(5) &\equiv c_{5,r} \end{aligned} \pmod{256}.$$

Thus the roots

$$\bar{z}_3 = 0 \quad \text{and} \quad \bar{z}_3 = 1$$

correspond respectively to the cyclotomic systems

$$\text{Cyc}_{0,k-1} \quad \text{and} \quad \text{Cyc}_{16,k-17}.$$

The relation

$$\bar{z}_3(\bar{z}_3 - \varepsilon_r) = 0$$

always permits $\text{Cyc}_{0,k-1}$, and permits $\text{Cyc}_{16,k-17}$ precisely when $\varepsilon_r = 1$. Consequently, the possibilities in (4.3) select exactly

$$\text{Cyc}_{0,k-1}$$

when $\varepsilon_r = 0$, and exactly

$$\text{Cyc}_{0,k-1}, \quad \text{Cyc}_{16,k-17}$$

when $\varepsilon_r = 1$.

4.4 Final classification

Proof of Theorem 1.5. Let Cyc_f be the cyclotomic packet selected in Section 4.3. Thus

$$a_3(f) \equiv_{\text{val}} \text{Cyc}_f(3) \pmod{256}, \quad a_5(f) \equiv_{\text{val}} \text{Cyc}_f(5) \pmod{256}.$$

Lemma 2.6 shows that Cyc_f defines a continuous weak character

$$\lambda_{\text{Cyc}_f} : \mathbb{T}_{\text{dc}}^{(2)} \longrightarrow \mathbf{Z}/256\mathbf{Z}$$

whose weight is congruent to k modulo 64. Consequently, λ_{Cyc_f} and the eigensystem of f have congruent values at T_3 , T_5 , and the weight operator [5]. Since

$$\mathbb{T}_{\text{dc}}^{(2)} = \overline{\mathbf{Z}_2[T_3, T_5, [5]]}$$

by Section 2.2.1, it follows that

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_f(n) \pmod{256} \quad ((n, 2) = 1).$$

Since $r \equiv k - 2 \pmod{16}$, the second packet is permitted precisely when $k - 2 \equiv 0, 6, 8, 14 \pmod{16}$, and is selected by the root $\bar{z}_3 = 1$. The two packets are distinct modulo 256, since their values at T_3 differ by 128.

Conversely, for each of the 48 permitted packets Cyc_f , the computation supplies a normalized cuspidal level-one strong eigenform g_{Cyc_f} of a weight

$$K_{\text{Cyc}_f} \equiv k \pmod{64}$$

whose T_3 - and T_5 -eigenvalues agree with $\text{Cyc}_f(3)$ and $\text{Cyc}_f(5)$ modulo 256. The weight congruence also gives agreement at [5]. The same big-Hecke generation argument therefore shows that the entire prime-to-2 eigensystem of g_{Cyc_f} is congruent to Cyc_f modulo 256. Consequently, every permitted packet occurs and the classification is exact. $\square$

5 The prime 3: classification modulo 27

We will call

$$r \equiv 2 \pmod{6} \quad \text{equivalently} \quad k \equiv 4 \pmod{6}$$

the *zero branch*, and

$$r \equiv 0, 4 \pmod{6} \quad \text{equivalently} \quad k \equiv 0, 2 \pmod{6}$$

the *nonzero branch*. As a consequence of the calculations below, these branches are also characterized by

$$\begin{aligned} \text{zero branch} &\iff a_2(f) \in 9\mathcal{O}_f, \\ \text{nonzero branch} &\iff a_2(f) \notin 9\mathcal{O}_f. \end{aligned}$$

For every degree residue r , put

$$t_r := 1 + 7^{r+1} \pmod{27}.$$

Thus

$$\begin{array}{c|cccccccc} r & 0 & 2 & 4 & 6 & 8 & 10 & 12 & 14 & 16 \\ \hline t_r & 8 & 20 & 14 & 17 & 2 & 23 & 26 & 11 & 5. \end{array} \tag{5.1}$$

For $r \in \{0, 4, 6, 10, 12, 16\}$, put $F_r(X) := X^2 - c_r$, where

$$\begin{array}{c|cccccc} r & 0 & 4 & 6 & 10 & 12 & 16 \\ \hline c_r & 1 & 4 & 4 & 1 & 7 & 7. \end{array} \tag{5.2}$$

5.1 Certified Hecke operator identities

5.1.1 The nonzero branch

Let

$$r \in \{0, 4, 6, 10, 12, 16\}.$$

We work at precision 27. The Dickson multipliers have degrees

$$a_3 = 54, \quad b_3 = 36, \quad \text{with } a_3 + b_3 = 90.$$

Moreover,

$$\chi_3(T_2) = 2^9 T_2 \equiv -T_2 \pmod{27}, \quad \chi_3(T_7) = 7^9 T_7 \equiv T_7 \pmod{27}.$$

For every degree

$$0 \leq d \leq 88, \quad d \equiv r \pmod{18},$$

and both orientations $q \in \{0, 1\}$, put

$$\mathcal{Z}_{2,q} := \frac{t_2''}{3} = \mathfrak{D}_{t_2,3}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/27\mathbf{Z})),$$

where t_2 acts as $\chi_3(2)^q T_2 = (-1)^q T_2$. The computation [Rus26, `classification_mod_27.ipynb`] certifies

$$\text{dom } \mathcal{Z}_{2,q} = \mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/27\mathbf{Z}), \quad F_r(\mathcal{Z}_{2,q}) \equiv 0 \pmod{9}.$$

It also certifies

$$(T_7 - t_{d \bmod 18}) \mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/27\mathbf{Z}) = 0. \quad (5.3)$$

By Proposition 3.12 and Theorem 3.8, these relations propagate to every even coefficient degree

$$d \geq 36, \quad d \equiv r \pmod{18}.$$

The remaining degrees

$$0 \leq d < 36, \quad d \equiv r \pmod{18},$$

are verified directly. Consequently, the relations hold in every even coefficient degree

$$d \equiv r \pmod{18}.$$

For such a degree, Proposition 3.11, with $m = 3$, $\alpha_1 = 1$, $b = 2$, and $S = 1$, gives

$$Z_2 := \frac{T_2}{3} \in \text{End}_{\mathbf{Z}_3}(P_d^+), \quad Z_2 L_d^+ \subseteq L_d^+,$$

and

$$(Z_2^2 - c_r) P_d^+ \subseteq 9P_d^+.$$

Similarly,

$$(T_7 - t_r) P_d^+ \subseteq 27P_d^+.$$

Consequently,

$$(Z_2^2 - c_r) L_d^+ \subseteq 9L_d^+, \quad (T_7 - t_r) L_d^+ \subseteq 27L_d^+. \quad (5.4)$$

5.1.2 The zero branch

Let

$$r \in \{2, 8, 14\}.$$

Put

$$G_2 := T_2(T_2 - 63), \quad Q(X) := (X(X + 1))^2.$$

We work at precision 243. The Dickson multipliers have degrees

$$a_5 = 486, \quad b_5 = 324, \quad a_5 + b_5 = 810,$$

and

$$\chi_5(T_2) = 2^{81}T_2 \equiv -T_2 \pmod{243}.$$

For every degree

$$326 \leq d \leq 806, \quad d \equiv 2 \pmod{6},$$

and both orientations $q \in \{0, 1\}$, put

$$\mathcal{H}_{2,q} := \frac{t_2(t_2 - 63)}{81} = \mathfrak{D}_{t_2(t_2 - 63), 81}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/243\mathbf{Z})),$$

where t_2 acts as $\chi_5(2)^q T_2 = (-1)^q T_2$. The computation [Rus26, `classification_mod_27.ipynb`] verifies

$$\text{dom } \mathcal{H}_{2,q} = \mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/243\mathbf{Z}), \quad Q(\mathcal{H}_{2,q}) \equiv 0 \pmod{3}.$$

Here

$$\chi_5^0(G_2) = T_2(T_2 - 63), \quad \chi_5^1(G_2) = T_2(T_2 + 63) \pmod{243}.$$

Proposition 3.12, followed by Proposition 3.11 with $m = 5$, $\alpha_1 = 4$, $b = 1$, and $S = 1$, therefore gives, for every even degree

$$d \geq 324, \quad d \equiv 2 \pmod{6},$$

an integral operator

$$H_2 := \frac{G_2}{81} = \frac{T_2(T_2 - 63)}{81} \in \text{End}_{\mathbf{Z}_3}(P_d^+), \quad H_2 L_d^+ \subseteq L_d^+. \quad (5.5)$$

It satisfies

$$Q(H_2)P_d^+ \subseteq 3P_d^+, \quad Q(H_2)L_d^+ \subseteq 3L_d^+. \quad (5.6)$$

The computation verifies (5.5) and (5.6) directly in the remaining 54 degrees

$$0 \leq d < 324, \quad d \equiv 2 \pmod{6}.$$

Consequently, both relations hold in every zero-branch degree.

Separately, for every degree

$$0 \leq d \leq 88, \quad d \equiv r \pmod{18},$$

and both orientations $q \in \{0, 1\}$, the precision-27 computation verifies

$$(T_7 - t_r)\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/27\mathbf{Z}) = 0. \quad (5.7)$$

Theorem 3.8 gives

$$(T_7 - t_r)P_d^+ \subseteq 27P_d^+, \quad (T_7 - t_r)L_d^+ \subseteq 27L_d^+$$

for every zero-branch degree

$$d \geq 36, \quad d \equiv r \pmod{18}.$$

The remaining zero-branch degrees

$$0 \leq d < 36, \quad d \equiv r \pmod{18},$$

are verified directly. Consequently, these two relations hold in every zero-branch degree

$$d \equiv r \pmod{18}.$$

Combining this with (5.6), we obtain

$$(H_2(H_2 + 1))^2 L_d^+ \subseteq 3L_d^+, \quad (T_7 - t_r)L_d^+ \subseteq 27L_d^+. \quad (5.8)$$

5.2 Congruences for a_2 and a_7

By the Eichler–Shimura comparison recalled in Section 3.3, we can choose a primitive simultaneous eigenvector

$$v_f \in L_d^+ \otimes_{\mathbf{Z}_3} \mathcal{O}_f$$

for the prime-to-3 Hecke eigensystem of f .

5.2.1 The nonzero branch

Suppose

$$r \in \{0, 4, 6, 10, 12, 16\}.$$

By (5.4),

$$a_7(f) \equiv_{\text{lit}} t_r \equiv_{\text{lit}} 1 + 7^{r+1} \pmod{27}. \quad (5.9)$$

Since

$$Z_2 = \frac{T_2}{3}$$

preserves L_d^+ , its eigenvalue z_2 on v_f is integral. Thus

$$z_2 := \frac{a_2(f)}{3} \in \mathcal{O}_f.$$

Again by (5.4),

$$z_2^2 \equiv_{\text{lit}} c_r \pmod{9}. \quad (5.10)$$

For each given c_r , we can find $\alpha \in \mathbf{Z}$ such that $\alpha^2 \equiv c_r \pmod{9}$ and $\alpha \not\equiv 0 \pmod{3}$. Since

$$(z_2 - \alpha)(z_2 + \alpha) \equiv_{\text{lit}} z_2^2 - c_r \equiv_{\text{lit}} 0 \pmod{9},$$

this means that either $z_2 - \alpha$ or $z_2 + \alpha$ is a unit in $\mathcal{O}_f$. Hence,

$$z_2 \in \pm\alpha + 9\mathcal{O}_f.$$

The roots modulo 9, together with the resulting values of $a_2(f)$ modulo 27, are

c_r	$z_2 \pmod{9}$	$a_2(f) \pmod{27}$
1	1, 8	3, 24
4	2, 7	6, 21
7	4, 5	12, 15.

(5.11)

5.2.2 The zero branch

Suppose

$$r \in \{2, 8, 14\}.$$

By (5.8),

$$a_7(f) \equiv_{\text{lit}} t_r \equiv_{\text{lit}} 1 + 7^{r+1} \pmod{27}. \quad (5.12)$$

By (5.5), the operator

$$H_2 = \frac{T_2(T_2 - 63)}{81}$$

preserves L_d^+ . Its eigenvalue on v_f is therefore integral:

$$h_2 := \frac{a_2(f)(a_2(f) - 63)}{81} \in \mathcal{O}_f. \quad (5.13)$$

Equivalently,

$$a_2(f)(a_2(f) - 63) \in 81\mathcal{O}_f.$$

We claim that

$$a_2(f) \in 9\mathcal{O}_f.$$

Indeed, if $v_3(a_2(f)) < 2$, then, since $v_3(63) = 2$,

$$v_3(a_2(f) - 63) = v_3(a_2(f)).$$

Consequently,

$$v_3(a_2(f)(a_2(f) - 63)) = 2v_3(a_2(f)) < 4,$$

contradicting

$$a_2(f)(a_2(f) - 63) \in 81\mathcal{O}_f.$$

Thus $v_3(a_2(f)) \geq 2$, as claimed.

Write

$$a_2(f) = 9x_2, \quad x_2 \in \mathcal{O}_f.$$

Then (5.13) becomes

$$h_2 = x_2(x_2 - 7).$$

Evaluating (5.6) on v_f gives

$$(h_2(h_2 + 1))^2 \in 3\mathcal{O}_f.$$

After reduction modulo $\mathfrak{m}_{\mathcal{O}_f}$, this gives

$$(\bar{x}_2(\bar{x}_2 - 1)(\bar{x}_2(\bar{x}_2 - 1) + 1))^2 = 0.$$

We obtain

$$\bar{x}_2 \in \{0, 1, -1\} \subseteq \mathbf{F}_3.$$

Hence

$$x_2 \equiv_{\text{val}} 0, 1, \text{ or } -1 \pmod{3}. \quad (5.14)$$

Consequently,

$$a_2(f) \equiv_{\text{val}} 0, 9, \text{ or } -9 \pmod{27}. \quad (5.15)$$

5.3 Packet selection

Recall that $r \equiv k-2 \pmod{18}$. Hence $r+1 \equiv k-1 \pmod{18}$. For each residue r , the possible values of $a_2(f)$ and $a_7(f)$ obtained above determine a unique unordered pair $\{a, b\} \subset \mathbf{Z}/18\mathbf{Z}$ satisfying

$$a + b \equiv r + 1 \equiv k - 1 \pmod{18}$$

such that

$$a_2(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(2) = 2^a + 2^b \pmod{27},$$

and

$$a_7(f) \equiv_{\text{lit}} \text{Cyc}_{a,b}(7) = 7^a + 7^b \pmod{27}.$$

On the nonzero branch, the congruence at T_2 is also literal modulo 27.

5.4 Final classification

Proof of Theorem 1.6. Let $\{a, b\} \subset \mathbf{Z}/18\mathbf{Z}$ be the unique pair selected above. Choose integers $i \geq 3$ and even $\kappa \geq 4$ such that

$$(a, b) \equiv (i, i + \kappa - 1) \pmod{18}.$$

By Lemma 2.6, we get a corresponding weak Hecke character

$$\lambda_{\kappa,i} : \mathbb{T}_{\text{dc}}^{(3)} \longrightarrow \mathbf{Z}/27\mathbf{Z}$$

of some weight K satisfying

$$K \equiv \kappa + 2i \pmod{18}.$$

Since

$$a + b \equiv k - 1 \pmod{18},$$

we have

$$K \equiv \kappa + 2i \equiv a + b + 1 \equiv k \pmod{18}.$$

Hence

$$4^K \equiv 4^k \pmod{27},$$

so the weak character and the character of f have the same $t_\Lambda = [4]$ -value.

By the packet selection above, their reductions modulo 27 (in the $\equiv_{\text{val}}$ sense) agree on T_2 and T_7 , and by the preceding weight congruence they agree on t_Λ . Therefore (2.4) gives

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(n) \pmod{27} \quad ((n, 3) = 1).$$

Conversely, the finite computation verifies that every cyclotomic packet appearing in the classification is realized by a strong level-one eigenform. Hence the classification is exact.

If $k \equiv 0, 2 \pmod{6}$, then f lies on the nonzero branch, where the agreement at T_2 and T_7 is literal. Repeating the same comparison in $\mathcal{O}_f/27\mathcal{O}_f$ gives the corresponding literal congruence. $\square$

6 The prime 5: classification modulo 25

For every even residue $r \pmod{20}$, define β_r by

$$\begin{array}{c|cccccccccc} r \pmod{20} & 0 & 2 & 4 & 6 & 8 & 10 & 12 & 14 & 16 & 18 \\ \hline \beta_r & 4 & 0 & 2 & 0 & 4 & 1 & 0 & 3 & 0 & 1. \end{array} \quad (6.1)$$

Put

$$Q_r(X) := (X - \beta_r)^3 + (X - \beta_r). \quad (6.2)$$

For $\eta \in \{0, 1\}$, define

$$F_{r,\eta}(X, Y) := X - \delta_{r,\eta}(c_r Y + 3 - \eta) - 5\gamma_{r,\eta}, \quad (6.3)$$

and put

$$G_r(X, Y) := F_{r,0}(X, Y)F_{r,1}(X, Y),$$

where

$$c_r := \begin{cases} 1, & r \equiv 0 \pmod{4}, \\ 2, & r \equiv 2 \pmod{4} \end{cases}$$

and the constants δ and γ are given by

r	$(\delta_{r,0}, \gamma_{r,0})$	$(\delta_{r,1}, \gamma_{r,1})$
0	(1, 4)	(4, 0)
2	(4, 4)	(1, 3)
4	(1, 1)	(4, 2)
6	(4, 2)	(1, 1)
8	(1, 3)	(4, 4)
10	(4, 0)	(1, 4)
12	(1, 0)	(4, 1)
14	(4, 3)	(1, 2)
16	(1, 2)	(4, 3)
18	(4, 1)	(1, 0).

6.1 Certified Hecke operator identities

We work at precision 25. The Dickson multipliers have degrees

$$a_2 = 100, \quad b_2 = 30, \quad \text{with } a_2 + b_2 = 130.$$

We also have

$$\chi_2(T_2) = 2^5 T_2 \equiv 7T_2, \quad \chi_2(T_{19}) = 19^5 T_{19} \equiv -T_{19} \pmod{25}.$$

Define the family

$$\mathcal{G}_{r,+} := \{G_r, \chi_2^2(G_r)\}, \quad \mathcal{G}_{r,-} := \{\chi_2(G_{r+10}), \chi_2^3(G_{r+10})\}$$

where the indices are understood modulo 20. The two transitions of Remark 3.9 are

$$(r, H, \epsilon) \longmapsto (r, H, \epsilon)$$

and

$$(r, H, \epsilon) \longmapsto (r + 10, \chi_2(H), -\epsilon).$$

The first plainly preserves the family. For the second,

$$\chi_2(\mathcal{G}_{r,+}) = \mathcal{G}_{r+10,-}, \quad \chi_2(\mathcal{G}_{r,-}) = \mathcal{G}_{r+10,+},$$

since $\chi_2^4 = \text{id}$. Thus the family $\mathcal{G}_{r,\epsilon}$ is closed under both transitions.

The exact induction base for Theorem 3.8 consists of the even degrees

$$30 \leq d \leq 128.$$

For each such d , put $r \equiv d \pmod{20}$. The computation [Rus26, `classification_mod_25.ipynb`] verifies

$$H\mathbb{M}_d^+(\mathbf{Z}/25\mathbf{Z}) = 0 \quad (H \in \mathcal{G}_{r,+})$$

and

$$H\mathbb{M}_d^-(\mathbf{Z}/25\mathbf{Z}) = 0 \quad (H \in \mathcal{G}_{r,-}).$$

Theorem 3.8, together with Remark 3.9, therefore propagates the relations to every even degree $d \geq 30$. The remaining even degrees $0 \leq d < 30$ are verified directly. In particular, for $r \equiv d \pmod{20}$,

$$G_r(T_{19}, T_2)L_d^+ \subseteq 25L_d^+. \quad (6.4)$$

For each even degree $30 \leq d \leq 128$ and each $q \in \{0, 1, 2, 3\}$, put

$$\mathcal{Z}_{19,q} := \frac{t_{19} - 5\beta_{d+10q}}{5} = \mathfrak{D}_{t_{19}-5\beta_{d+10q}, 5}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/25\mathbf{Z})),$$

where the subscript of β is understood modulo 20 and t_{19} acts as $\chi_2(19)^q T_{19} = (-1)^q T_{19}$. Writing $F(X) := (X^3 + X)^2$, the computation verifies

$$\text{dom } \mathcal{Z}_{19,q} = \mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/25\mathbf{Z}), \quad F(\mathcal{Z}_{19,q}) \equiv 0 \pmod{5}.$$

These four conditions are closed under the two transitions of Remark 3.9. Indeed, the A_2 -transition preserves the residue, the sign, and the orientation, while the B_2 -transition sends

$$(d, q) \mapsto (d - 30, q + 1)$$

and reverses the sign. Since

$$(d - 30) + 10(q + 1) \equiv d + 10q \pmod{20},$$

the subscript of β is unchanged. Thus the defining presentation is preserved, while the change in orientation accounts for the twist of T_{19} . Hence the same strong-induction argument as in Proposition 3.12 propagates these conditions to every even degree $d \geq 30$. By Proposition 3.11, with $m = 2$, $\alpha_1 = 1$, $b = 1$, and $S = 1$, we obtain, for $r \equiv d \pmod{20}$,

$$\frac{T_{19} - 5\beta_r}{5} \in \text{End}_{\mathbf{Z}_5}(P_d^+).$$

Consequently,

$$Z_{19} := \frac{T_{19}}{5} \in \text{End}_{\mathbf{Z}_5}(P_d^+), \quad Z_{19}L_d^+ \subseteq L_d^+,$$

and

$$Q_r(Z_{19})^2 P_d^+ \subseteq 5P_d^+.$$

By saturation,

$$Q_r(Z_{19})^2 L_d^+ \subseteq 5L_d^+.$$

The remaining even degrees $0 \leq d < 30$ are verified directly. Thus, in every even degree,

$$Q_r(Z_{19})^2 P_d^+ \subseteq 5P_d^+, \quad Q_r(Z_{19})^2 L_d^+ \subseteq 5L_d^+. \quad (6.5)$$

6.2 Congruences for a_2 and a_{19}

By the Eichler–Shimura comparison recalled in Section 3.3, choose a primitive simultaneous eigenvector

$$v_f \in L_d^+ \otimes_{\mathbf{Z}_5} \mathcal{O}_f$$

for the prime-to-5 Hecke eigensystem of f .

Since $Z_{19} = T_{19}/5$ preserves L_d^+ , its eigenvalue z_{19} on v_f is integral. Thus

$$z_{19} := \frac{a_{19}(f)}{5} \in \mathcal{O}_f.$$

Equation (6.5) gives

$$Q_r(z_{19})^2 \in 5\mathcal{O}_f.$$

After reduction to the residue field,

$$Q_r(\bar{z}_{19}) = 0.$$

Since

$$Q_r(X) \equiv (X - \beta_r)(X - \beta_r - 2)(X - \beta_r - 3) \pmod{5},$$

there is a unique

$$\lambda \in \{\beta_r, \beta_r + 2, \beta_r + 3\}$$

such that

$$\bar{z}_{19} = \bar{\lambda}.$$

Equivalently,

$$a_{19}(f) \equiv_{\text{val}} 5\lambda \pmod{25}. \quad (6.6)$$

Put

$$x_f := c_r a_2(f) + 3.$$

Evaluating (6.4) on the primitive eigenvector v_f gives

$$(a_{19}(f) - \delta_{r,0}x_f - 5\gamma_{r,0})(a_{19}(f) - \delta_{r,1}(x_f - 1) - 5\gamma_{r,1}) \in 25\mathcal{O}_f. \quad (6.7)$$

Reducing modulo the maximal ideal of $\mathcal{O}_f$, and using $a_{19}(f) \in 5\mathcal{O}_f$, we obtain

$$(-\delta_{r,0}\bar{x}_f)(-\delta_{r,1}(\bar{x}_f - 1)) = 0.$$

Since

$$\delta_{r,0}, \delta_{r,1} \in \mathbf{F}_5^\times,$$

there is a unique $\epsilon \in \{0, 1\}$ such that

$$\bar{x}_f = \epsilon. \quad (6.8)$$

For this ϵ , the factor corresponding to $1 - \epsilon$ in (6.7) is a unit. Hence

$$a_{19}(f) - \delta_{r,\epsilon}(x_f - \epsilon) - 5\gamma_{r,\epsilon} \in 25\mathcal{O}_f. \quad (6.9)$$

Together with (6.6), this gives

$$x_f \equiv_{\text{val}} \epsilon + 5\delta_{r,\epsilon}^{-1}(\lambda - \gamma_{r,\epsilon}) \pmod{25}. \quad (6.10)$$

Since $x_f = c_r a_2(f) + 3$ and $c_r \in \{1, 2\}$, we therefore obtain

$$a_2(f) \equiv_{\text{val}} c_r^{-1} \left(\epsilon - 3 + 5\delta_{r,\epsilon}^{-1}(\lambda - \gamma_{r,\epsilon}) \right) \pmod{25}. \quad (6.11)$$

6.3 Packet selection

Recall that

$$d \equiv r \equiv k - 2 \pmod{20}.$$

For each of the six pairs (ϵ, λ) selected by (6.6) and (6.8), a direct calculation of the cyclotomic packets gives a unique unordered pair

$$\{a, b\} \subset \mathbf{Z}/20\mathbf{Z}, \quad a + b \equiv k - 1 \pmod{20},$$

such that

$$a_2(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(2) \pmod{25}, \quad a_{19}(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(19) \pmod{25}.$$

Moreover, the pair can be written

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/20\mathbf{Z},$$

where

$$2i + \kappa \equiv k \pmod{20}$$

and

$$\kappa \in \begin{cases} \{10, 14, 22\}, & k \equiv 0, 2 \pmod{10}, \\ \{6, 14, 18\}, & k \equiv 4, 8 \pmod{10}, \\ \{6, 14, 22\}, & k \equiv 6 \pmod{10}. \end{cases} \quad (6.12)$$

Thus there are exactly six possible cyclotomic packets for each residue class of k modulo 20, and hence 60 packets over all weights.

6.4 Final classification

Proof of Theorem 1.7. Let

$$\{a, b\} \subset \mathbf{Z}/20\mathbf{Z}$$

be the unique pair selected in Section 6.3. Thus, in every weight, we have a unique pair satisfying

$$a + b \equiv k - 1 \pmod{20},$$

which can be written

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/20\mathbf{Z},$$

with

$$\kappa \in \begin{cases} \{10, 14, 22\}, & k \equiv 0, 2 \pmod{10}, \\ \{6, 14, 18\}, & k \equiv 4, 8 \pmod{10}, \\ \{6, 14, 22\}, & k \equiv 6 \pmod{10}, \end{cases}$$

and

$$2i + \kappa \equiv k \pmod{20}.$$

Choosing the representative of i sufficiently large, we may assume $i \geq 2$. Moreover,

$$a_2(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(2) \pmod{25}, \quad a_{19}(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(19) \pmod{25}.$$

By Lemma 2.6, this packet is realized by a continuous weak character

$$\lambda_{\kappa,i} : \mathbb{T}_{\text{dc}}^{(5)} \longrightarrow \mathbf{Z}/25\mathbf{Z}$$

of some weight K satisfying

$$K \equiv \kappa + 2i \equiv k \pmod{20}.$$

Hence

$$6^K \equiv 6^k \pmod{25},$$

so $\lambda_{\kappa,i}$ and the Hecke character of f have the same $t_\Lambda = [6]$ -value in the valuative coefficient quotient.

By the packet selection, the two characters also agree there on T_2 and T_{19} . By (2.6), they agree on the whole completed prime-to-5 Hecke algebra. Therefore

$$a_n(f) \equiv_{\text{val}} \text{Cyc}_{a,b}(n) \pmod{25} \quad ((n, 5) = 1).$$

Finally, since $20 = \varphi(25)$ and $i \geq 2$, the equivalence following Definition 1.4 gives

$$\theta^{20} f \equiv_{\text{val}} \theta^i G_\kappa \pmod{25}.$$

Conversely, the finite computation verifies that each of the 60 cyclotomic packets appearing above is realized by a strong level-one eigenform. Hence the classification is exact. $\square$

7 The prime 7: classification modulo 49

7.1 Certified Hecke operator identities

Define

$$\begin{aligned} F_0(X) &:= X^6 + 7X^5 + 45X^4 + 7X^3 + 4X^2 + 7X, \\ F_2(X) &:= X^6 + 42X^5 + 6X^4 + 9X^2 + 14X, \\ F_4(X) &:= X^6 + 7X^5 + 47X^4 + 14X^3 + X^2 + 28X. \end{aligned} \tag{7.1}$$

Modulo 7, these factor as

$$\begin{aligned} F_0(X) &\equiv (X(X-3)(X-4))^2, \\ F_2(X) &\equiv (X(X-2)(X-5))^2, \pmod{7}, \\ F_4(X) &\equiv (X(X-1)(X-6))^2 \end{aligned} \tag{7.2}$$

Also put

$$\{\alpha_j, \beta_j\} = \begin{cases} \{3, 4\}, & j = 0, \\ \{2, 5\}, & j = 2, \\ \{1, 6\}, & j = 4, \end{cases} \quad \mathcal{C}_j := \{0, \alpha_j, \beta_j\}.$$

Define

$$Q_j(X) := \frac{F_j(X)}{49}, \quad G_j(X, Y) := \frac{Y - 2 - X(X - \alpha_j)(X - \beta_j)}{7}, \quad j \in \{0, 2, 4\}. \tag{7.3}$$

Finally,

$$H(X) := (X^7 - X)^3.$$

We first treat $Q_{r \bmod 6}(T_3)$. The computation is performed at precision $343 = 7^3$. The Dickson multipliers have degrees

$$a_3 = 2058, \quad b_3 = 392, \quad \text{with} \quad a_3 + b_3 = 2450,$$

so the exact induction base consists of the even degrees

$$392 \leq d < 2450.$$

For every such d and every $0 \leq q < 6$, put

$$\mathcal{Q}_q := \frac{F_{(d+2q) \bmod 6}(t_3)''}{49} = \mathfrak{D}_{F_{(d+2q) \bmod 6}(t_3), 49}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/343\mathbf{Z})), \quad \epsilon_q = (-1)^q,$$

where t_3 acts as $\chi_3(3)^q T_3$. The computation [Rus26, classification_mod_49.ipynb] verifies

$$\text{dom } \mathcal{Q}_q = \mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/343\mathbf{Z}), \quad H(\mathcal{Q}_q) \equiv 0 \pmod{7}.$$

These six conditions are closed under both transitions of Remark 3.9: the A_3 -transition preserves $d + 2q \pmod{6}$, while the B_3 -transition sends

$$(d, q) \mapsto (d - 392, q + 1)$$

and again preserves $d + 2q \pmod{6}$.

Now fix an even residue $r \bmod 42$. The preceding transition compatibility and the same strong-induction argument as in Proposition 3.12 propagate these conditions to every degree

$$d \geq 392, \quad d \equiv r \pmod{42}.$$

Applying Proposition 3.11 in orientation $q = 0$, with $m = 3$, $\alpha_1 = 2$, $b = 1$, and $S = 1$, gives

$$Q_{r \bmod 6}(T_3) \in \text{End}_{\mathbf{Z}_7}(P_d^+), \quad Q_{r \bmod 6}(T_3)L_d^+ \subseteq L_d^+.$$

Moreover,

$$H(Q_{r \bmod 6}(T_3))P_d^+ \subseteq 7P_d^+, \quad H(Q_{r \bmod 6}(T_3))L_d^+ \subseteq 7L_d^+. \quad (7.4)$$

The remaining degrees

$$0 \leq d < 392, \quad d \equiv r \pmod{42},$$

are verified directly. Consequently, these conclusions hold in every even degree

$$d \equiv r \pmod{42}.$$

We next treat $G_{r \bmod 6}(T_3, T_{29})$. This computation is performed at precision 49. The Dickson multipliers have degrees

$$a_2 = 294, \quad b_2 = 56, \quad \text{with} \quad a_2 + b_2 = 350,$$

so the exact induction base consists of the even degrees

$$56 \leq d < 350.$$

For every such d and every $0 \leq q < 6$, put

$$\mathcal{G}_q := \frac{7G_{(d+2q) \bmod 6}(t_3, t_{29})''}{7} = \mathfrak{D}_{7G_{(d+2q) \bmod 6}(t_3, t_{29}), 7}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/49\mathbf{Z})), \quad \epsilon_q = (-1)^q,$$

where t_n acts as $\chi_2(n)^q T_n$. Here $7G_j$ denotes the integral polynomial

$$7G_j(X, Y) = Y - 2 - X(X - \alpha_j)(X - \beta_j).$$

The computation verifies

$$\text{dom } \mathcal{G}_q = \mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/49\mathbf{Z}), \quad H(\mathcal{G}_q) \equiv 0 \pmod{7}.$$

These six conditions are closed under both transitions of Remark 3.9. Hence the same strong-induction argument as in Proposition 3.12 propagates these conditions to every degree

$$d \geq 56, \quad d \equiv r \pmod{42}.$$

Applying Proposition 3.11 in orientation $q = 0$, with $m = 2$, $\alpha_1 = 1$, $b = 1$, and $S = 1$, gives

$$G_{r \bmod 6}(T_3, T_{29}) \in \text{End}_{\mathbf{Z}_7}(P_d^+), \quad G_{r \bmod 6}(T_3, T_{29})L_d^+ \subseteq L_d^+.$$

Moreover,

$$H(G_{r \bmod 6}(T_3, T_{29}))P_d^+ \subseteq 7P_d^+, \quad H(G_{r \bmod 6}(T_3, T_{29}))L_d^+ \subseteq 7L_d^+. \quad (7.5)$$

The remaining degrees

$$0 \leq d < 56, \quad d \equiv r \pmod{42},$$

are verified directly. Consequently, these conclusions hold in every even degree

$$d \equiv r \pmod{42}.$$

7.2 Congruences for a_3 and a_{29}

By the Eichler–Shimura comparison recalled in Section 3.3, choose a primitive simultaneous eigenvector

$$v_f \in L_d^+ \otimes_{\mathbf{Z}_7} \mathcal{O}_f$$

for the prime-to-7 Hecke eigensystem of f .

We first determine $a_3(f)$ modulo 7. Since

$$Q_{r \bmod 6}(T_3) = \frac{F_{r \bmod 6}(T_3)}{49}$$

is integral on L_d^+ , evaluation on v_f gives

$$F_{r \bmod 6}(a_3(f)) \in 49\mathcal{O}_f.$$

Reducing modulo the maximal ideal and using (7.2), there is a unique

$$c \in \mathcal{C}_{r \bmod 6}$$

such that

$$a_3(f) \equiv c \pmod{\mathfrak{m}_{\mathcal{O}_f}}.$$

Write $a_3(f) = c + y$. Direct calculation gives

$$F_{r \bmod 6}(c) \in 49\mathbf{Z}_7, \quad F'_{r \bmod 6}(c) \in 7\mathbf{Z}_7, \quad \frac{F''_{r \bmod 6}(c)}{2} \in \mathbf{Z}_7^\times.$$

If $0 < v_7(y) < 1$, then in the Taylor expansion of $F_{r \bmod 6}(c + y)$ the quadratic term has strictly smaller valuation than every other term, contradicting

$$F_{r \bmod 6}(a_3(f)) \in 49\mathcal{O}_f.$$

It follows that

$$a_3(f) - c \in 7\mathcal{O}_f. \quad (7.6)$$

Since $G_{r \bmod 6}(T_3, T_{29})$ is also integral on L_d^+ , evaluation on v_f gives

$$a_{29}(f) - 2 - a_3(f)(a_3(f) - \alpha_{r \bmod 6})(a_3(f) - \beta_{r \bmod 6}) \in 7\mathcal{O}_f.$$

By (7.6), one of the three factors in the cubic product lies in $7\mathcal{O}_f$. Hence

$$a_{29}(f) - 2 \in 7\mathcal{O}_f. \quad (7.7)$$

We may therefore write

$$a_3(f) = c + 7x_f, \quad a_{29}(f) = 2 + 7y_f, \quad x_f, y_f \in \mathcal{O}_f. \quad (7.8)$$

Put

$$q_f := Q_{r \bmod 6}(a_3(f)), \quad g_f := G_{r \bmod 6}(a_3(f), a_{29}(f)).$$

By the preceding subsection,

$$q_f, g_f \in \mathcal{O}_f$$

and

$$H(q_f), H(g_f) \in 7\mathcal{O}_f.$$

Reducing modulo the maximal ideal gives

$$\bar{q}_f^7 = \bar{q}_f, \quad \bar{g}_f^7 = \bar{g}_f,$$

so

$$\bar{q}_f, \bar{g}_f \in \mathbf{F}_7.$$

Plugging

$$a_3 = c + 7x_f, \quad a_{29} = 2 + 7y_f, \quad x_f, y_f \in \mathcal{O}_f,$$

into (7.3) and expanding, we find that

$r \bmod 6$	c	$\bar{q}_f$	$\bar{g}_f$
0	0	$4\bar{x}_f^2 + \bar{x}_f$	$\bar{y}_f - 5\bar{x}_f$
0	3	$2\bar{x}_f^2 + 2\bar{x}_f + 3$	$\bar{y}_f - 4\bar{x}_f$
0	4	$2\bar{x}_f^2$	$\bar{y}_f - 4\bar{x}_f$
2	0	$2\bar{x}_f^2 + 2\bar{x}_f$	$\bar{y}_f - 3\bar{x}_f$
2	2	$\bar{x}_f^2 + 3\bar{x}_f + 4$	$\bar{y}_f - \bar{x}_f$
2	5	$\bar{x}_f^2 + 4\bar{x}_f$	$\bar{y}_f - \bar{x}_f$
4	0	$\bar{x}_f^2 + 4\bar{x}_f$	$\bar{y}_f - 6\bar{x}_f$
4	1	$4\bar{x}_f^2 + \bar{x}_f + 2$	$\bar{y}_f - 2\bar{x}_f$
4	6	$4\bar{x}_f^2 + 2\bar{x}_f + 5$	$\bar{y}_f - 2\bar{x}_f$

(7.9)

Since $\bar{q}_f, \bar{g}_f \in \mathbf{F}_7$, we see that $\bar{x}_f, \bar{y}_f \in \mathbf{F}_{49}$. Thus these relations are already enough to prove the finiteness of the number of away-from-7 Hecke eigensystems in level 1.

In order to obtain classification, we have to correlate $\bar{q}_f$ and $\bar{g}_f$. For $c \in \mathcal{C}_j$, let $E_{j,c}(X) \in \mathbf{Z}[X]$ be the coefficientwise lift, with coefficients in $\{0, \dots, 6\}$, of the Lagrange polynomial

$$\prod_{c' \in \mathcal{C}_j \setminus \{c\}} \frac{X - c'}{c - c'} \in \mathbf{F}_7[X].$$

The denominators are nonzero in $\mathbf{F}_7$, since the elements of $\mathcal{C}_j$ are distinct modulo 7. Thus

$$E_{j,c}(a_3(f)) \equiv \begin{cases} 1 & (\text{mod } \mathfrak{m}_{\mathcal{O}_f}), \quad a_3(f) \equiv c \pmod{\mathfrak{m}_{\mathcal{O}_f}}, \\ 0 & (\text{mod } \mathfrak{m}_{\mathcal{O}_f}), \quad a_3(f) \equiv c' \pmod{\mathfrak{m}_{\mathcal{O}_f}} \text{ for some } c' \in \mathcal{C}_j \setminus \{c\}. \end{cases}$$

The classification computation is performed at precision 343. For every even degree

$$392 \leq d < 2450$$

and every orientation $0 \leq q < 6$, put

$$r \equiv d + 14q \pmod{42}.$$

On

$$\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/343\mathbf{Z}), \quad \epsilon_q = (-1)^q,$$

the computation supplies polynomials

$$R_{r,c}^{(1)}, R_{r,c}^{(2)} \in \mathbf{Z}_7[X, Y]$$

for every $c \in \mathcal{C}_{r \bmod 6}$. We use the same notation for their reductions in $\mathbf{F}_7[X, Y]$. On this source module, put

$$\begin{aligned} \mathcal{Q}_q &= " \frac{F_{r \bmod 6}(t_3)}{49} " &:= \mathfrak{D}_{F_{r \bmod 6}(t_3), 49}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/343\mathbf{Z})), \\ \mathcal{G}_q &= " \frac{7G_{r \bmod 6}(t_3, t_{29})}{7} " &:= \mathfrak{D}_{7G_{r \bmod 6}(t_3, t_{29}), 7}(\mathbb{M}_d^{\epsilon_q}(\mathbf{Z}/343\mathbf{Z})), \end{aligned}$$

where t_n acts as $\chi_3(n)^q T_n$. Thus both relations are now taken at precision 343. Let

$$S_{c,q} := E_{r \bmod 6,c}(\chi_3(3)^q T_3)^6.$$

For $\nu \in \{1, 2\}$, the computation verifies

$$[(R_{r,c}^{(\nu)})^3](\mathcal{Q}_q, \mathcal{G}_q) \Gamma_{S_{c,q}} \equiv 0 \pmod{7}.$$

Here the polynomial is cubed and expanded before evaluating it at the linear relations.

These conditions form a periodic family closed under both transitions of Remark 3.9. Indeed, the A_3 -transition preserves r , while the B_3 -transition sends

$$(d, q) \mapsto (d - 392, q + 1)$$

and preserves r modulo 42, since

$$-392 + 14 = -378 \equiv 0 \pmod{42}.$$

Consequently, Proposition 3.12, followed in orientation $q = 0$ by Proposition 3.11 with $m = 3$, division exponents 2, 1, $b = 1$, and $S = E_{r \bmod 6,c}(T_3)^6$, gives, for every even degree $d \geq 392$ with $d \equiv r \pmod{42}$,

$$E_{r \bmod 6,c}(T_3)^6 \left[R_{r,c}^{(\nu)}(Q_{r \bmod 6}(T_3), G_{r \bmod 6}(T_3, T_{29})) \right]^3 L_d^+ \subseteq 7L_d^+.$$

The remaining even degrees $0 \leq d < 392$ are verified directly.

We illustrate the argument on the zero branch $c = 0$ in coefficient-degree residue $r = 0$. Since $E_{0,0}(a_3(f))$ is a unit, it may be cancelled after evaluating the certified operator relations on v_f . After reduction modulo $\mathfrak{m}_{\mathcal{O}_f}$, the resulting joint relations give

$$\bar{q}_f(\bar{q}_f - 3)(\bar{q}_f - 4)(\bar{q}_f - 5) = 0, \quad \bar{g}_f = 2\bar{q}_f.$$

Now using the expressions for $\bar{q}_f$ and $\bar{g}_f$ in terms of $\bar{x}_f$ and $\bar{y}_f$ given in (7.9), we obtain

$$(\bar{x}_f, \bar{y}_f) \in \{(0, 0), (1, 1), (2, 4), (3, 2), (4, 2), (5, 4), (6, 1)\}.$$

In particular, all possible values of $\bar{x}_f$ and $\bar{y}_f$ belong to $\mathbf{F}_7$, even when the residue field of $\mathcal{O}_f$ is larger. Therefore, on this branch,

$$(a_3(f), a_{29}(f)) \equiv_{\text{val}} (7x, 2 + 7y) \pmod{49}$$

for exactly the seven pairs (x, y) listed above.

The computation verifies the analogous calculation for every even residue $r \pmod{42}$ with $c = 0$. In each case, the common zero set of the two coordinate relations pulls back under (7.9) to exactly seven pairs

$$(\bar{x}_f, \bar{y}_f) \in \mathbf{F}_7^2.$$

These pairs are enough for exact classification on the zero branch.

For the nonzero branches (i.e. $a_3 \equiv_{\text{lit}} c \pmod{7}$, $c \neq 0$), the relations

$$E_{r \bmod 6, c}(T_3)^6 \left[R_{r, c}^{(\nu)}(Q_{r \bmod 6}(T_3), G_{r \bmod 6}(T_3, T_{29})) \right]^3 L_d^+ \subseteq 7L_d^+$$

still determine the possible pairs $(\bar{q}_f, \bar{g}_f)$, but they do not by themselves determine the correct $(\bar{x}_f, \bar{y}_f)$ corresponding to strong eigenforms. They define a set

$$\mathcal{T}_{r, c} := \left\{ (q, g) \in \mathbf{F}_7^2 : R_{r, c}^{(1)}(q, g) = R_{r, c}^{(2)}(q, g) = 0 \right\}.$$

For every nonzero $c \in \mathcal{C}_{r \bmod 6}$, put formally

$$U_c := \frac{T_3 - c}{7}, \quad V := \frac{T_{29} - 2}{7}.$$

Here these expressions are used inside the staged division relations stated below; we do not require U_c or V separately to preserve L_d^+ .

On the same exact induction base and in the same six orientations as above, the computation verifies compatible staged division relations which propagate to every even degree

$$d \geq 392, \quad d \equiv r \pmod{42},$$

and give

$$E_{r \bmod 6, c}(T_3)^6 \left[E_{r \bmod 6, c}(T_3)^6 U_c - \phi_{r, c}(Q_{r \bmod 6}(T_3), G_{r \bmod 6}(T_3, T_{29})) \right]^3 L_d^+ \subseteq 7L_d^+$$

and

$$E_{r \bmod 6, c}(T_3)^6 \left[E_{r \bmod 6, c}(T_3)^6 V - \psi_{r, c}(Q_{r \bmod 6}(T_3), G_{r \bmod 6}(T_3, T_{29})) \right]^3 L_d^+ \subseteq 7L_d^+,$$

for some affine polynomials

$$\phi_{r,c}, \psi_{r,c} \in \mathbf{Z}_7[X, Y].$$

We use the same notation for their reductions in $\mathbf{F}_7[X, Y]$. The remaining even degrees $0 \leq d < 392$ are verified directly.

To justify these lattice inclusions, put $S_c := E_{r \bmod 6, c}(T_3)^6$. For $N = T_3 - c$ or $T_{29} - 2$, the first witness equation for the monomial $(S_c N/7)^3$, starting at $S_c u$, gives

$$S_c^2 N P_d^+ \subseteq 7 P_d^+.$$

By the integrality of $Q_{r \bmod 6}(T_3)$ and (7.2), S_c acts on $P_d^+/7P_d^+$ invertibly on the generalized c -eigenspace and as zero on the other generalized eigenspaces. Hence $\ker S_c^2 = \ker S_c$, so $S_c N P_d^+ \subseteq 7 P_d^+$. Thus $S_c U_c$ and $S_c V$ preserve P_d^+ , and also L_d^+ by saturation. Remark ??, applied with starting vectors $S_c u$ and division exponents 2, 1, 1, now gives the stated inclusions with retained precision $7^{3-2} = 7$.

After evaluation on v_f , put $e_f := E_{r \bmod 6, c}(a_3(f))$. Since $e_f \equiv 1 \pmod{\mathfrak{m}_{\mathcal{O}_f}}$, we may cancel the outer unit factor e_f^6 . Reducing modulo $\mathfrak{m}_{\mathcal{O}_f}$, the inner factors e_f^6 also reduce to 1, giving

$$\bar{x}_f = \phi_{r,c}(\bar{q}_f, \bar{g}_f), \quad \bar{y}_f = \psi_{r,c}(\bar{q}_f, \bar{g}_f).$$

Thus every point of $\mathcal{T}_{r,c}$ determines exactly one permitted pair $(\bar{x}_f, \bar{y}_f)$.

For example, take $r = 40$ and $c = 1$. The coordinate relations give

$$\mathcal{T}_{40,1} = \{(0, 1), (2, 0), (2, 3), (6, 6)\}.$$

The corresponding graph relations reduce to

$$\bar{x}_f = 3\bar{g}_f + 5, \quad \bar{y}_f = 3.$$

Evaluating these formulas on the four points of $\mathcal{T}_{40,1}$ gives

$$(\bar{x}_f, \bar{y}_f) \in \{(0, 3), (1, 3), (2, 3), (5, 3)\}.$$

Consequently, on this branch,

$$(a_3(f), a_{29}(f)) \equiv_{\text{val}} (1 + 7x, 2 + 7y) \pmod{49}$$

for precisely these four pairs (x, y) .

The other nonzero branch is treated in the same way. In every coefficient-degree residue, the centre-zero branch contributes seven signatures and each of the two nonzero branches contributes four, giving fifteen signatures in total. All of these signatures have coordinates in $\mathbf{Z}/49\mathbf{Z}$.

7.3 Packet selection

By the preceding subsection,

$$a_3(f) \equiv c \pmod{7\mathcal{O}_f}, \quad a_{29}(f) \equiv 2 \pmod{7\mathcal{O}_f}$$

for a unique

$$c \in \mathcal{C}_{r \bmod 6}.$$

There is a unique unordered pair

$$\{a, b\} \subset \mathbf{Z}/6\mathbf{Z}, \quad a + b \equiv k - 1 \pmod{6},$$

such that

$$c = 3^a + 3^b \quad \text{and} \quad 2 = 29^a + 29^b \quad \text{in } \mathbf{F}_7.$$

Explicitly,

$r \bmod 6$	c	$\{a, b\}$
0	0	$\{2, 5\}$
0	3	$\{3, 4\}$
0	4	$\{0, 1\}$
2	0	$\{0, 3\}$
2	2	$\{4, 5\}$
2	5	$\{1, 2\}$
4	0	$\{1, 4\}$
4	1	$\{2, 3\}$
4	6	$\{0, 5\}$

Thus the residue class $r \bmod 6$, together with the value $c \equiv a_3(f) \pmod{7}$, determines the residual cyclotomic packet of f . Equivalently, the pair may be written

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/6\mathbf{Z},$$

for suitable

$$i \in \{1, \dots, 6\}, \quad \kappa \in \{4, 8\}, \quad \kappa + 2i \equiv k \pmod{6}.$$

7.4 Final classification

Proof of Theorem 1.8. By the preceding subsections, every normalized cuspidal level-one eigenform f determines a unique signature

$$\sigma = (r, c, x, y), \quad r \equiv k - 2 \pmod{42}, \quad c \in \mathcal{C}_{r \bmod 6}, \quad x, y \in \mathbf{Z}/7\mathbf{Z},$$

such that

$$a_3(f) \equiv_{\text{val}} c + 7x \pmod{49}, \quad a_{29}(f) \equiv_{\text{val}} 2 + 7y \pmod{49}.$$

For each residue $r \bmod 42$, there are exactly 15 such signatures, and hence 315 signatures altogether.

The finite computation verifies that every one of these 315 signatures is realized by a normalized cuspidal level-one eigenform g_σ of weight $k_\sigma \leq 380$, with

$$k_\sigma - 2 \equiv r \pmod{42}.$$

If f has signature σ , then

$$a_3(f) \equiv_{\text{val}} a_3(g_\sigma) \pmod{49}, \quad a_{29}(f) \equiv_{\text{val}} a_{29}(g_\sigma) \pmod{49},$$

and

$$k_\sigma \equiv k \pmod{42}.$$

Since 3 has order 42 modulo 49, the latter congruence gives

$$3^{k_\sigma} \equiv 3^k \pmod{49},$$

so f and g_σ also have the same value of the weight operator $[3]$ modulo 49.

By Section 2.2.3,

$$\mathbb{T}_{\text{dc}}^{(7)} = \overline{\mathbf{Z}_7[T_3, T_{29}, [3]]}.$$

Hence the prime-to-7 Hecke eigensystems of f and g_σ are congruent modulo 49 in the valuative sense. Therefore

$$a_n(f) \equiv_{\text{val}} a_n(g_\sigma) \pmod{49} \quad ((n, 7) = 1).$$

Consequently, only finitely many prime-to-7 Hecke eigensystems occur modulo 49, and every such eigensystem is realized in weight at most 380. In fact, the argument gives exactly 315 eigensystems.

It remains to record their reduction modulo 7. By Section 7.3, there is a unique unordered pair

$$\{a, b\} \subset \mathbf{Z}/6\mathbf{Z}, \quad a + b \equiv k - 1 \pmod{6},$$

such that

$$a_3(f) \equiv_{\text{lit}} \text{Cyc}_{a,b}(3) \pmod{7}, \quad a_{29}(f) \equiv_{\text{lit}} \text{Cyc}_{a,b}(29) \pmod{7}.$$

Choose

$$i \in \{1, \dots, 6\}, \quad \kappa \in \{4, 8\},$$

such that

$$\{a, b\} = \{i, i + \kappa - 1\} \quad \text{in } \mathbf{Z}/6\mathbf{Z}.$$

Then

$$\kappa + 2i \equiv k \pmod{6}.$$

By Lemma 2.6, this cyclotomic packet is realized by a continuous weak character of $\mathbb{T}_{\text{dc}}^{(7)}$ whose weight is congruent to k modulo 6. It therefore has the same $[3]$ -value as f modulo 7, and by packet selection it also has the same T_3 - and T_{29} -values. The completed-Hecke generation above then gives

$$a_n(f) \equiv_{\text{lit}} \text{Cyc}_{a,b}(n) \pmod{7} \quad ((n, 7) = 1).$$

Equivalently,

$$\theta^6 f \equiv_{\text{lit}} \theta^i G_\kappa \pmod{7}.$$

This proves Theorem 1.8. □

Part III

Applications

8 Strong prime-to- p weight bounds

In [KRW16, §1.5], the finiteness conjecture for strong eigenforms modulo p^m is reformulated in terms of a strong weight bound: for fixed N, p, m , one asks for a constant $B(N, p, m)$

such that every strong eigenform of level N is congruent modulo p^m to a strong eigenform of the same level and of weight at most $B(N, p, m)$.

The classifications proved above concern only the prime-to- p Hecke eigensystem. We therefore introduce the corresponding prime-to- p notion. A number B is called a *strong prime-to- p weight bound* for (N, p, m) if every normalized characteristic-zero cuspidal eigenform f of level N is congruent modulo p^m , at all Hecke operators prime to p , to a normalized characteristic-zero cuspidal eigenform g of level N and weight at most B . We denote the least such bound, when it exists, by

$$B^{(p)}(N, p, m).$$

It is also natural to allow cyclotomic twists. A number B is called a *strong weight bound up to θ -twist* for (N, p, m) if, for every normalized characteristic-zero cuspidal eigenform f of level N , there exist an integer $i \geq 0$ and a normalized characteristic-zero cuspidal eigenform g of level N and weight at most B such that

$$a_n(f) \equiv n^i a_n(g) \pmod{p^m} \quad ((n, p) = 1).$$

We denote the least such bound, when it exists, by

$$B_\theta(N, p, m).$$

Equivalently, the prime-to- p eigensystem of f is congruent modulo p^m to the θ^i -twist of that of g .

The proved classifications above and the fixed-weight computations give the following bounds. The unmarked entries are proved and optimal; the entries marked $\dagger$ are conjectural.

The proved and conjectural values in Table 1 display some suggestive patterns. For $p = 5$ and $p = 7$, the proved values at $m = 1, 2$ satisfy

$$B^{(p)}(1, p, m) = (p + 2)\varphi(p^m) + 2.$$

At the higher displayed precisions, a different uniform pattern appears: the values for

$$(p, m) = (3, 3), (3, 4), (3, 5), (5, 3)$$

satisfy

$$B^{(p)}(1, p, m) = (p + 1)\varphi(p^m) - 2.$$

Among these four values, the case $(p, m) = (3, 3)$ is proved, while the other three are conjectural. The data for $p = 2$ appear less regular, although the bounds remain of the same order of magnitude as $\varphi(2^m)$. The bounds up to θ -twist likewise do not yet suggest a uniform formula valid for all primes and exponents. It seems premature to conjecture an exact general formula from these data, though it seems more reasonable to ask the following.

Question. Does there exist, for every prime p , a constant $C_p > 0$ such that

$$B^{(p)}(1, p, m) \leq C_p \varphi(p^m)$$

for every $m \geq 1$?

modulus	strong weight bound B	bound up to θ -twist B_θ
2	12	12
2^2	12	12
2^3	12	12
2^4	18	12
2^5	22	16
2^6	30	24
2^7	46	32
2^8	90	62
2^9	$158^\dagger$	$128^\dagger$
3	12	12
3^2	32	16
3^3	70	24
3^4	$214^\dagger$	$58^\dagger$
3^5	$646^\dagger$	$166^\dagger$
5	30	12
5^2	142	24
5^3	$598^\dagger$	$106^\dagger$
7	56	16
7^2	380	78

Table 1: Strong level-one weight bounds for prime-power away-from- p congruences. The bound B_θ means that every strong eigensystem is congruent away from p to $\theta^i g$ for some $i \geq 0$ and some strong level-one eigenform g of weight at most B_θ . The entries marked $\dagger$ are conjectural.

A Signatures modulo 49

For each even weight residue $k \bmod 42$, the permitted signatures are listed below. The entries c, x, y are taken in $\{0, \dots, 6\}$, and represent the congruences

$$a_3(f) \equiv_{\text{val}} c + 7x \pmod{49}, \quad a_{29}(f) \equiv_{\text{val}} 2 + 7y \pmod{49}.$$

Table 2: The 315 permitted signatures modulo 49.

$k \bmod 42$	c	Permitted pairs (x, y)
0	0	$\{(0, 6), (1, 1), (2, 0), (3, 3), (4, 3), (5, 0), (6, 1)\}$
	1	$\{(0, 3), (1, 3), (2, 3), (5, 3)\}$
	6	$\{(1, 3), (4, 3), (5, 3), (6, 3)\}$
2	0	$\{(0, 0), (1, 1), (2, 4), (3, 2), (4, 2), (5, 4), (6, 1)\}$
	3	$\{(0, 4), (1, 4), (3, 4), (6, 4)\}$
	4	$\{(0, 4), (3, 4), (5, 4), (6, 4)\}$
4	0	$\{(0, 4), (1, 1), (2, 6), (3, 5), (4, 5), (5, 6), (6, 1)\}$
	2	$\{(0, 5), (3, 5), (5, 5), (6, 5)\}$
	5	$\{(0, 5), (1, 5), (3, 5), (6, 5)\}$
6	0	$\{(0, 4), (1, 6), (2, 5), (3, 1), (4, 1), (5, 5), (6, 6)\}$
	1	$\{(0, 6), (1, 6), (4, 6), (6, 6)\}$
	6	$\{(0, 6), (2, 6), (5, 6), (6, 6)\}$
8	0	$\{(0, 0), (1, 1), (2, 4), (3, 2), (4, 2), (5, 4), (6, 1)\}$
	3	$\{(0, 0), (2, 0), (5, 0), (6, 0)\}$
	4	$\{(0, 0), (1, 0), (4, 0), (6, 0)\}$
10	0	$\{(0, 6), (1, 3), (2, 1), (3, 0), (4, 0), (5, 1), (6, 3)\}$
	2	$\{(1, 1), (2, 1), (3, 1), (6, 1)\}$
	5	$\{(0, 1), (3, 1), (4, 1), (5, 1)\}$
12	0	$\{(0, 1), (1, 3), (2, 2), (3, 5), (4, 5), (5, 2), (6, 3)\}$
	1	$\{(0, 2), (1, 2), (2, 2), (5, 2)\}$
	6	$\{(1, 2), (4, 2), (5, 2), (6, 2)\}$
14	0	$\{(0, 6), (1, 0), (2, 3), (3, 1), (4, 1), (5, 3), (6, 0)\}$
	3	$\{(0, 3), (3, 3), (4, 3), (5, 3)\}$
	4	$\{(1, 3), (2, 3), (3, 3), (6, 3)\}$
16	0	$\{(0, 0), (1, 4), (2, 2), (3, 1), (4, 1), (5, 2), (6, 4)\}$
	2	$\{(1, 4), (2, 4), (3, 4), (6, 4)\}$
	5	$\{(0, 4), (3, 4), (4, 4), (5, 4)\}$
18	0	$\{(0, 4), (1, 6), (2, 5), (3, 1), (4, 1), (5, 5), (6, 6)\}$
	1	$\{(1, 5), (3, 5), (4, 5), (5, 5)\}$
	6	$\{(1, 5), (2, 5), (3, 5), (5, 5)\}$
20	0	$\{(0, 4), (1, 5), (2, 1), (3, 6), (4, 6), (5, 1), (6, 5)\}$
	3	$\{(0, 6), (1, 6), (2, 6), (4, 6)\}$
	4	$\{(2, 6), (4, 6), (5, 6), (6, 6)\}$

Continued on the next page

Table 2 (continued)

$k \bmod 42$	c	Permitted pairs (x, y)
22	0	$\{(0, 0), (1, 4), (2, 2), (3, 1), (4, 1), (5, 2), (6, 4)\}$
	2	$\{(0, 0), (3, 0), (5, 0), (6, 0)\}$
	5	$\{(0, 0), (1, 0), (3, 0), (6, 0)\}$
24	0	$\{(0, 6), (1, 1), (2, 0), (3, 3), (4, 3), (5, 0), (6, 1)\}$
	1	$\{(1, 1), (2, 1), (3, 1), (6, 1)\}$
	6	$\{(0, 1), (3, 1), (4, 1), (5, 1)\}$
26	0	$\{(0, 1), (1, 2), (2, 5), (3, 3), (4, 3), (5, 5), (6, 2)\}$
	3	$\{(0, 2), (3, 2), (4, 2), (5, 2)\}$
	4	$\{(1, 2), (2, 2), (3, 2), (6, 2)\}$
28	0	$\{(0, 6), (1, 3), (2, 1), (3, 0), (4, 0), (5, 1), (6, 3)\}$
	2	$\{(0, 3), (1, 3), (4, 3), (6, 3)\}$
	5	$\{(0, 3), (2, 3), (5, 3), (6, 3)\}$
30	0	$\{(0, 0), (1, 2), (2, 1), (3, 4), (4, 4), (5, 1), (6, 2)\}$
	1	$\{(1, 4), (2, 4), (3, 4), (6, 4)\}$
	6	$\{(0, 4), (3, 4), (4, 4), (5, 4)\}$
32	0	$\{(0, 4), (1, 5), (2, 1), (3, 6), (4, 6), (5, 1), (6, 5)\}$
	3	$\{(0, 5), (2, 5), (5, 5), (6, 5)\}$
	4	$\{(0, 5), (1, 5), (4, 5), (6, 5)\}$
34	0	$\{(0, 4), (1, 1), (2, 6), (3, 5), (4, 5), (5, 6), (6, 1)\}$
	2	$\{(2, 6), (4, 6), (5, 6), (6, 6)\}$
	5	$\{(0, 6), (1, 6), (2, 6), (4, 6)\}$
36	0	$\{(0, 0), (1, 2), (2, 1), (3, 4), (4, 4), (5, 1), (6, 2)\}$
	1	$\{(1, 0), (3, 0), (4, 0), (5, 0)\}$
	6	$\{(1, 0), (2, 0), (3, 0), (5, 0)\}$
38	0	$\{(0, 6), (1, 0), (2, 3), (3, 1), (4, 1), (5, 3), (6, 0)\}$
	3	$\{(0, 1), (1, 1), (3, 1), (6, 1)\}$
	4	$\{(0, 1), (3, 1), (5, 1), (6, 1)\}$
40	0	$\{(0, 1), (1, 5), (2, 3), (3, 2), (4, 2), (5, 3), (6, 5)\}$
	2	$\{(0, 2), (1, 2), (4, 2), (6, 2)\}$
	5	$\{(0, 2), (2, 2), (5, 2), (6, 2)\}$

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